Label granularity in supervised pretraining of jet foundation models

[author list]

Abstract

Foundation models for particle physics are often pretrained to classify simulated jets, the sprays of particles from quarks and gluons, into a growing number of classes. Fine-grained classes let a single model serve any coarser signal definition, but whether they also improve the learned representation is untested with the training data fixed. We retrain the Sophon model, a Particle Transformer variant pretrained on the JetClass-II dataset, with its 188 classes and merged label sets of 162, 43 and 17 classes, three seeds each; models with the same seed share their training examples, in order, and all initial weights outside the output layer. Controls are random partitions of the classes, some with no or a single class boundary between decays that differ only in quark flavor. Linear probes are scored by $1 - \mathrm{AUC}$ (AUC: area under the receiver operating characteristic curve). On JetClass-II, merging the classes of background jets raises $1 - \mathrm{AUC}$ only on background jets, by a factor of 1.153 ± 0.035, while removing quark flavor from the labels raises it over all classes by a further factor of 1.47 ± 0.04. Between decays into $b\bar{b}$ and $c\bar{c}$, $1 - \mathrm{AUC}$ rises by a factor of 4.06 ± 0.35 at 17 classes but only 1.55 ± 0.15 for random partitions into 17 classes, which keep apart many other decays that differ only in quark flavor. On JetClass-II, the flavor boundaries of a label set therefore matter more than its number of classes; on JetClass, the random partitions are no better than the 17-class label set. On top and quark/gluon tagging, frozen features gain little over those of a randomly initialized network. On the full training sets, fine-tuning improves on training from scratch on both, and the label set changes its result by about a per cent. **[fine-tuning on JetClass-II and JetClass; $X \to b\bar{b}$ and $X \to bs$ tagging]**

1 Introduction

Quarks and gluons produced in proton–proton collisions at the Large Hadron Collider (LHC) [28] are observed as jets, collimated sprays of hadrons that

the ATLAS [2] and CMS [18] detectors measure particle by particle. When a heavy particle—a top quark, a W, Z or Higgs boson, or a hypothetical new particle—is produced with large momentum, its decay products merge into a single jet of large radius whose internal structure reflects the decay [40, 35]. Identifying the particle that initiated a jet, jet tagging, is central to measurements of the Higgs boson and to searches for new particles [20, 21]. Much of it is flavor tagging: hadrons that contain a b quark live about 1.5 ps and those with a c quark 1 ps or less, so they decay a measurable distance from the collision and leave displaced tracks and secondary vertices [19, Sec. 4.1]. Current taggers are deep networks that read the measured particles of a jet [53, 54, 5, 23], trained on simulated jets whose origin the simulation records.

Jet foundation models pretrain one network on a large simulated dataset and adapt it to many tasks [54, 44, 49, 9, 29]. Among the pretraining objectives compared so far, classifying the simulated jets is the strongest single objective once enough labeled jets are available [27]. Because the simulation records the full decay behind every jet, its classes can be made as fine as one likes, and they have grown. Sophon [44], built on the Particle Transformer (ParT) [54], is pretrained on the JetClass-II dataset [45] with 188 classes: 161 decays of a hypothetical resonance X, labeled by their decay products, and 27 classes of jets from quantum chromodynamics (QCD) multijet production, the main background of such searches, labeled by their numbers of b, c and s quarks. The CMS tagger GloParT grew from 37 to 374 classes between its first and third versions [24].

Fine classes have an established benefit at the output of the model. For an ideally trained classifier, the score of a merged class is the sum of the scores of the classes it contains, and adding classes leaves the ratios between the scores of the original classes unchanged [44, Properties 1–2, Eqs. (2)–(3)]; one fine-grained model therefore serves any coarser definition of the signal. These properties concern the output of an ideal classifier. They say nothing about which features a network of finite size learns, and the learned features are what is reused when the model is adapted to a new task. Whether fine classes improve them has not been tested with the training data held fixed. Sophon's 188-class model tagged $X \to b\bar{b}$ decays better than a 42-class model [44, Sec. III], but that model saw 2.5 million examples per epoch instead of 10 million, and only QCD jets and two-prong decays [44, App. B.2]; its 42 classes are unrelated to the 43-class level studied here. In the GloParT study, the classes added to a classifier came with their training jets [43, Sec. 11.4.1]. Controlled studies of jet foundation models have varied

the training objective [27, 55, 16], the composition of the training data[1] [59], and the model and dataset size [60, 61], but not the label set with the data fixed.

Computer vision suggests that the answer is not obvious. Pretraining on 127 groups of the 1000 ImageNet classes cost 2.8 points of detection precision after fine-tuning and about 15% in nearest-neighbor classification of the fine classes on the frozen features [33], and on a taxonomy of species, finer pretraining labels helped only up to about 10^3 classes [32]. Natural images, however, carry labels only at the granularities that annotators provide; simulated jets can be labeled exactly at every level of a class hierarchy, and the same jets can be used at every level.

We retrain Sophon on JetClass-II with its 188 classes and with 162, 43 and 17 classes, four levels of one hierarchy in which each level merges classes of the one before, with three seeds each. Models with the same seed see the same training jets in the same order from the same initial weights, so two of them differ only in their labels and their output layer (Fig. 1). Random partitions of the classes into 17 groups, with the group shares of the 17-class level, separate the number of classes from what the classes distinguish, and partitions with a controlled set of flavor boundaries test which distinctions matter. A mass-regression output, as used by CMS taggers [23, 43], is added at 162 and 17 classes, and 188- and 17-class models are also pretrained on the kinematic inputs alone, the only inputs the top-tagging benchmark provides. Each model is evaluated with probes on its frozen features and with full fine-tuning: on the classification of JetClass-II into its 188 classes, on JetClass [54], on top tagging [14] and quark/gluon tagging [37], on pairs of classes that the label sets merge, on the jet mass, and on the $X \to b\bar{b}$ and $X \to bs$ benchmark of Sophon.

[summary of findings, written from the final results]

Section 2 places the study among related work, Section 3 describes the design, Section 4 gives the results and Section 5 discusses them.

2 Related work

Classification as the pretraining objective. Supervised pretraining set the reference for jet foundation models: pretraining ParT on JetClass raised its top-tagging background rejection at 30% signal efficiency from 1602 to 2766 [54, Table 5], and OmniLearn [49] and OmniLearned [9] fine-

[1] Ref. [59] changes which JetClass-II classes enter pretraining at a fixed number of jets, so the jets change together with the labels.

tune one pretrained model on many tasks. Self-supervised objectives have been studied in detail—masked particle modeling [29, 41], generative pretraining [11], contrastive learning on resimulated jets [31] and joint-embedding prediction [6, 42]—and their gain is largest with few labels. Masked particle modeling beat training from scratch on JetClass with fewer than 10k labeled jets [29], and in a comparison of objectives at fixed data it beat classification only for the smallest model fine-tuned on 10^3 to 10^4 top jets, while classification was best on the full 1.2×10^6 jets [27, Table XVII]. Since classification is the strongest single objective when labels are plentiful, its label set deserves a controlled study. The uncertainties of these studies come from repeated downstream training of a single pretrained model [54, 29], and Sophon reports a single model [44]; we pretrain every configuration several times, because the variation between pretraining runs sets the scale against which a difference between label sets has to be judged [57].

Label granularity in vision. Training on fine labels raises the accuracy on coarse classes of the same images, but the gain becomes negligible as the coarse classes grow in number, 0.16 and 0.01 points at 15 and 20 coarse classes of CIFAR-100 [17, Table VIII]. The same study also regrouped whole CIFAR-10 classes at random into two coarse classes of fixed size, and fine labels still helped, by up to 2.78 points [17, Table VI]: the nearest precedent for our random partitions, which differ in two ways—they have 17 groups, and they are pretraining label sets evaluated on fixed downstream tasks rather than targets in their own right. Whether fine labels help depends on how the classes are grouped: in small networks they helped with small training sets and hurt with large ones, never helped for a taxonomic grouping of four CIFAR-10 classes (cats and dogs against cars and trucks), and always helped for a mixed grouping of the same classes [52, Figs. 2–3]. With about a billion web images, pretraining on a larger hashtag vocabulary raised the accuracy of a linear classifier on the frozen features by about 7 points on the ImageNet task with the most classes [48, Sec. 3.1.2]; after full fine-tuning, the accuracies on that task were 48.5% and 48.4%, and the smaller, ImageNet-aligned vocabulary won on the 1000 classes of ImageNet, 84.2% against 83.6% [48, Fig. 1]. The images changed together with the vocabulary. Random class labels assigned to individual images gave a higher error than any level of a species taxonomy [32, Table 1]; our random partitions instead regroup whole decay classes, so every label remains a function of the decay.

What coarse labels remove. Training on coarse labels can remove variation within a class that later tasks need. On the BREEDS benchmark of living things [56], a classifier on frozen features that saw one labeled image of each fine class reached 33.83% after pretraining on the coarse classes and 58.95% after pretraining on the fine classes [13, Table 2]. Checkpoints whose features varied less within each superclass transferred worse even after full fine-tuning [34, Sec. 4], and methods that learn fine-grained features from coarse labels add an objective that tells individual images apart [63, 58, 13]. With the labels fixed, losses that separate the training classes more strongly give frozen features that transfer worse (rank correlation -0.93), while after full fine-tuning no loss is significantly better than the softmax cross-entropy [38]. Frozen-feature probes and full fine-tuning can therefore disagree, and we report both.

Auxiliary targets. An auxiliary objective can help or hurt a tagger. Removing both auxiliary objectives of the ATLAS flavor tagger GN2, which predict the origin of each track and whether two tracks share a vertex, reduced its c- and light-jet rejection by up to 30% [5, Discussion]. CMS taggers based on ParT add a jet-mass regression to the classification loss: the published tagger regresses the mass in GeV with weight 0.05 [23], and GloParT weights its two mass-regression terms by 5 [43, Eq. (11.7)]. Ref. [43] argues that, because the terms do not interact, the minimum of the combined loss is optimal for both tasks [43, Sec. 11.2.1], and that the best regression depends on the regression targets of all classes [43, Sec. 11.3.2]. ATLAS, in contrast, found a combined network for boosted Higgs bosons about 10% less efficient than a dedicated classifier and regresses the mass in a separate network [4, Sec. 6.5]. Whether the effect of a mass target on the learned features depends on the label set has not been measured.

Transfer to new signals. Sophon has no class for $X \to bs$ decays and tags them with a classifier on its frozen features whose first layer is initialized from the classification head; this classifier outperformed ParT and ParticleNet trained on the task, while the discriminant of the $X \to bq$ output performed significantly worse [44, Sec. III].

3 Methods

3.1 Data, model and pretraining

JetClass-II [44, 45] contains about 139 million simulated jets clustered with the anti-k_{T} algorithm [15] at $R = 0.8$. Resonances are generated with MadGraph5_aMC@NLO [1] and QCD multijet events with Pythia [10], which also simulates decays, the parton shower and hadronization; the detector is simulated with Delphes [26] in a CMS-like configuration, with an average of 50 additional proton–proton interactions per event suppressed by PUPPI [8]. In a two-prong decay, X decays directly into a pair of quarks, gluons or leptons; in three- and four-prong decays, it decays into a pair of intermediate resonances Y. The masses of X and Y are sampled for each generated event, so a class is defined by its decay products and not by a mass. We select jets with $200 < p_{\mathrm{T}} < 2{,}500$ GeV and $20 < m_{\mathrm{SD}} < 500$ GeV, where m_{SD} is the soft-drop mass [39] with $z_{\mathrm{cut}} = 0.1$ and $\beta = 0$ [22]; every JetClass-II jet also satisfies Sophon's third requirement, $|\eta| < 2.5$ [44, App. A]. As in Sophon, the training jets are reweighted to the same distribution in $(p_{\mathrm{T}}, m_{\mathrm{SD}})$ for each of Sophon's reweighting classes, to avoid sculpting the background mass distribution when a selection is applied to the score [44, Sec. II]; the reweighting is the same for every label set.

The network is Sophon: the ParT architecture, with a pairwise bias in the attention of its 8 particle-attention blocks, 2 class-attention blocks and embedding dimension 128, a classification head with a hidden layer of 512 units, and particle momenta rescaled so that every jet has the same transverse momentum [44, Sec. II]. The number of particle-attention blocks follows the released code [46]; the paper states 6, but its 2.3 million parameters correspond to 8. Every model is pretrained for 80 epochs of 10,240,000 jets, as in Sophon's released configuration [46], with the Ranger optimizer [64, 47] and a learning rate that is constant for the first 70% of training and then decays exponentially (Appendix A). All models with the same seed are trained on the same examples in the same order and start from the same initial weights except for the output layer, whose width is the number of classes. The initial weights follow PyTorch's default initialization rather than that of Sophon's code, a difference shared by every label set. We evaluate the checkpoint with the highest validation accuracy among epochs 70–79, where the learning rate is lowest, and read its representation at the input of the head: the 128-dimensional class token after the class-attention blocks and the final layer normalization.

Table 1: Published discriminants built from sums of Sophon's output scores, against the number of classes in the label set (columns, finest first). •: the discriminant can be built exactly from that label set's outputs; —: it cannot. This is a property of the label set (Section 3.2), not of a trained model.

discriminant	source	188	162	64	43	30	17	4	2
$X \to b\bar{b}$ vs QCD	[44], Eq. (4)	•	•	•	—	—	—	—	—
$X \to c\bar{s}$ vs QCD (Sophon's A_1)	[44], Eqs. (5)–(6)	•	•	—	—	—	—	—	—
three quarks, exactly one b, vs QCD (Sophon's A_3)	[44], Eqs. (5)–(6)	•	•	—	—	—	—	—	—
$W' \to W\phi \to WWW$ event discriminant	[44], Eqs. (7)–(9)	•	•	—	—	—	—	—	—
D_{bc}, boosted $\lvert V_{cb}\rvert$ measurement	[65], Eq. (1)	•	•	—	—	—	—	—	—

Pretrained in this study: the 188-, 162-, 43- and 17-class label sets.

3.2 Label sets

Class hierarchy. The four levels (Fig. 1a) remove quark flavor before decay structure, so that the coarsest still describes the decay topology. The 188-class level is Sophon's. The 162-class level merges the 27 QCD classes into one and keeps every resonant class. The 43-class level, reached through a level of 64 classes, labels each decay by its topology—its pattern of intermediate resonances and prongs—and by whether it contains a b quark, else a c quark, else only light quarks (u, d, s) or gluons; it therefore merges gluons with light quarks, electrons with muons, and decays that differ only in their numbers of b or c quarks. Treating a gluon as a light parton is our choice; Sophon's classes keep the two apart. The 17-class level removes quark flavor and keeps, for each decay, its number of prongs and the type of each: quark or gluon, electron or muon, τ lepton decaying to hadrons or to an electron or muon, or neutrino. At every level a class contains only QCD jets, only decays into quarks and gluons, or only decays with leptons. Figure 2 shows the classes of selected decays at each level, and Table 1 which of the published discriminants built from Sophon's output scores each level can form.

Random partitions. A boundary between classes that keeps apart two decays differing only in quark flavor, such as $X \to b\bar{b}$ and $X \to c\bar{c}$, is a *flavor boundary*. The 17-class level has no flavor boundary and groups decays by topology. Each of the five random partitions instead assigns the resonant classes at random to 17 groups, one of them for all QCD jets, each resonant group holding the same share of the reweighted training jets as a group of the 17-class level (within 5%), so that it carries the same weight in the loss. Of 32 draws per partition, each refined by a search

that reduces the number of class pairs it merges that the 17-class level also merges, the one with the fewest such pairs is kept; the partitions are thus built to differ from the topological grouping as much as the shares allow, and a group can mix decays with different numbers of prongs. They have many flavor boundaries. Three further partitions, designed after the results of the hierarchy, fix where the flavor boundaries lie (Appendix B). The partition with no flavor boundary places all decays that differ only in quark flavor in the same group. The partition with a b-quark boundary moves, from it, the four-quark decays $X \to YY \to QQQQ$ that contain a b quark into another group, with a compensating move of other decays to keep the shares; its only flavor boundary lies between the four-quark decays with and without a b quark, and it keeps 8 of the 11 pairs of four-quark decays that differ by exchanging b and c quarks apart. The partition with a random flavor boundary makes the same compensating move but moves four-quark decays chosen at random, keeping none of those pairs apart. The random partitions are pretrained with one seed each, the partitions with controlled flavor boundaries with two seeds each.

Mass regression. The 162- and 17-class models are also pretrained, with two seeds each, with an output that regresses

$$y = \ln\bigl(m_{\mathrm{SD}}^{\mathrm{gen}}/m_{\mathrm{SD}}^{\mathrm{reco}}\bigr), \tag{1}$$

the logarithm of the ratio between the soft-drop mass of the matched generator-level jet and that of the reconstructed jet, for jets with a generator-level match. The target, its log-cosh loss and its weight $\lambda = 5.0$ relative to the classification loss are our choices, fixed before training. The precedents differ: GloParT regresses generator-level masses directly with a log-cosh loss and weights its two regression terms by 5, a value it tuned [43, Eqs. (11.2)–(11.7)], and the published CMS tagger regresses the mass in GeV with weight 0.05 [23]. The weight is the same at both label sets, so the share of the training loss it gives the mass term is not: averaged over training, it is 13.78 and 13.87% in the 162-class runs and 28.65 and 28.67% in the 17-class runs, because the classification loss is smaller with fewer classes. The comparison between the label sets is therefore made at a fixed weight, not at a matched share.

Kinematic inputs only. The top-tagging benchmark provides only the four-momenta of the particles, from which 7 of the 17 input features are computed. The 188- and 17-class levels are therefore also pretrained, with

one seed, on these features alone, with the same jets, order and recipe; their initial weights are shared with each other but not with the models on all features, whose first layer is wider.

Randomly initialized network. The saved initial weights of each 188-class run give a randomly initialized network with the same architecture, evaluated by the same probes as the pretrained models. A second version has its batch-normalization statistics recomputed on training jets, so that a comparison with it does not rest on the scaling of the inputs alone.

3.3 Evaluation

Tasks. Every model is scored on the same training and test jets. The first task is the classification of JetClass-II into its 188 classes, on jets from files that no pretraining run read. The training, validation and test sets hold equal numbers of jets of each class, as JetClass does: up to 5,000 test jets of each class, all of a class's jets where it has fewer ([census] QCD classes, the rarest with [census] jets in the whole test split), and training sets of 10^3 to 10^6 jets with an equal share of each class (Appendix C). These classes are the pretraining labels of the 188-class models, so the task favors finer label sets by construction; we report the average over all classes and over the resonant and the QCD classes separately, since only the 188-class level divides the QCD jets. The second task is the 10-class classification of JetClass [54], which has all 17 input features but different classes and a different simulation, without pileup: the in-distribution test of whether a difference carries over. The third and fourth are the public benchmarks of top tagging [14], separating hadronic top-quark decays from QCD jets, and quark/gluon tagging, separating jets of light quarks from jets of gluons, on the Pythia sample of Ref. [37] as released [36]: 2 million jets at $R = 0.4$ with $500 < p_{\mathrm{T}} < 550$ GeV and $|y| < 1.7$, generated without a detector simulation, split 1.6/0.2/0.2 million as recommended by ParT [54, Sec. 5.2]. The quark/gluon models are also scored on the companion sample generated with Herwig 7.1.4 [7, 51].

Inputs the benchmarks lack. The network reads 17 features per particle. The top-tagging data lack the charge, the five particle-type flags and the four track-displacement variables; the quark/gluon data lack the displacement variables and carry the particle type as five binary flags and the charge, ParT's *exp* scenario [54, Table 2]. Missing features are set to zero. In JetClass-II every particle has exactly one flag set and only neutral particles

have zero displacement, so these zeros are inputs no pretraining jet has. The field avoids this mismatch in three ways: ParT pretrains a separate model on the features each benchmark provides [54, Sec. 5.2]; OmniLearn zeroes the particle-identification inputs at random, with probability 0.2, during pretraining [49, Sec. II.A]; and OmniLearned encodes particle identification and vertexing in separate blocks that contribute zero when absent, and pretrains on a pooled set of datasets from several detectors [9]. We follow ParT for top tagging only, with the kinematic-input models. For the comparison of label sets on top and quark/gluon tagging, every model receives the same inputs and the same zeros, and what the zeros cost is measured on JetClass-II itself: the features a benchmark lacks are set to zero in both the training and the test jets of the probe, or, as a check that separates the lost information from the unfamiliar input, replaced by values drawn from their JetClass-II distribution—one particle-type flag drawn with the JetClass-II frequencies, a matching charge, and the displacement of a random JetClass-II particle of the same charge.

Metrics. On JetClass-II and JetClass we report the accuracy and the macro-averaged area under the receiver operating characteristic curve (AUC), the mean one-versus-rest AUC over the classes, on every test jet. On JetClass we also give ParT's one-versus-one average [54] and the background rejection $1/\epsilon_B$ of each class against QCD jets at ParT's working points; on JetClass-II, each class's AUC and rejection against QCD jets. On top and quark/gluon tagging we report the accuracy, the AUC and the rejection at signal efficiencies of 50% and 30%, R_{50} and R_{30}. Label sets are compared by the ratio of $1 - \text{AUC}$, the fraction of signal–background pairs ranked in the wrong order, because AUC values close to one hide large differences: with pretraining, ParT's top-tagging AUC rises only from 0.9858 to 0.9877, while its rejection at 30% signal efficiency rises from 1602 to 2766 [54, Table 5].

Probes. A linear probe is a multinomial logistic regression on the standardized class token, its regularization chosen on validation jets (Appendix C), fitted on training sets of 10^3 jets up to the largest of each task. Because a linear probe reads only what a linear function of the features can, every conclusion that rests on a match between configurations or on the randomly initialized network is checked with a nonlinear probe, a multilayer perceptron with one hidden layer of 256 units, at the largest training set of each task. Sophon's transfer classifier starts from the hidden layer of the head; for the 188-class models we also probe this 512-unit layer.

Pairs of classes. Binary probes measure distinctions that the label sets merge at different levels. The resonant pairs are $X \to bc$ against $X \to bq$ and $X \to cs$—the two-prong classes that enter the denominator of the discriminant proposed in Ref. [65] for a measurement of $|V_{cb}|$ with boosted $W \to bc$ jets—and $X \to b\bar{b}$ against $X \to c\bar{c}$ and $X \to YY \to bbqq$ against $X \to YY \to ccqq$. Each is merged in some random partitions and kept apart in others. The QCD pairs, which only the 188-class level distinguishes, are the bb against the cc class, and QCD jets with a b quark, and with a c but no b quark, each against QCD jets with no b, c or s quark. Each side holds up to 10^5 training and 2×10^5 test jets at its natural composition.

Fine-tuning. Each pretrained model is fine-tuned once from its checkpoint, end to end, with a newly initialized head; the same network trained from random initialization, with three seeds, is the reference. On the full training sets of top and quark/gluon tagging we use ParT's fine-tuning recipe [54, Sec. 5.2], 20 epochs at constant learning rates of 10^{-4} for the pretrained weights and 50 times that for the head, and ParT's from-scratch recipe for the reference. Below the full set, 20 epochs of a small training set amount to a few hundred optimizer steps, far from convergence, so every run—pretrained or from scratch—gets the same budget: at most 20,000 steps, a validation pass every 500 steps on a fixed validation set, and the best checkpoint, stopping once 10 evaluations bring no improvement. The learning rate of the pretrained weights, **[sweep]**, and of the network from scratch, **[sweep]**, were chosen on the validation sets of top and quark/gluon tagging at 10^4 jets (Appendix C). Every pretrained model is fine-tuned on top tagging, quark/gluon tagging and JetClass; on JetClass-II, the levels of the hierarchy and the models with the mass regression are.

Resonance tagging. We follow Sophon's benchmark [44, Sec. III]: a spin-0 resonance of **[samples not in]** GeV decaying to $b\bar{b}$ or to bs. Sophon's background was a sample of Standard Model processes after a trigger and a leading-jet selection, and its figure of merit the discovery significance at a fixed luminosity. We use the QCD jets of the JetClass-II test files, without trigger and without jets from W, Z or top quarks, and report the signal efficiency at a fixed QCD efficiency and $\epsilon_S/\sqrt{\epsilon_B}$, which is proportional to the significance when the signal is small [25]; the values rank the models on a common background and are not comparable with Sophon's. All scores are evaluated in Sophon's soft-drop mass window, $150 < m_{\mathrm{SD}} < 230$ GeV for a resonance of 200 GeV, scaled by the ratio of the resonance masses.

$X \to b\bar{b}$ is tagged with Sophon's discriminant [44, Eq. (4)] by the label sets that contain it, and by every label set with a linear probe and Sophon's transfer classifier on the frozen features; $X \to bs$, which no label set contains, only with the latter two. As in Sophon, the discriminant of the class that contains $X \to bq$ shows what the output scores give without training, and the network trained from scratch on the same jets what training without pretraining reaches (Appendix C).

Mass resolution and sculpting. The target y of Eq. (1) is evaluated on the JetClass-II test jets with a generator-level match, split into training, validation and evaluation parts. The reference is the uncorrected soft-drop mass; for the models with the mass regression, its output is the corrected mass, and for every model a small multilayer perceptron also regresses y from the frozen features, after subtracting the median of each JetClass-II class so that a better classifier is not credited as a better mass regressor. The resolution is σ_{eff}, half the smallest interval containing 68% of the residuals $\ln(m_{\text{reg}}/m_{\text{SD}}^{\text{gen}})$, following Ref. [43, Sec. 8.2.2], and the response is the median of $m_{\text{reg}}/m_{\text{SD}}^{\text{gen}}$, as ATLAS reports them as functions of p_{T} [3, Sec. 6]. Mass sculpting is measured as CMS does [23]: a cut on $1 - p_{\text{QCD}}$, the summed probability of the non-QCD classes, at QCD efficiencies of 10% and 1%, and the Jensen–Shannon distance between the soft-drop mass distributions of the accepted and of all QCD jets.

3.4 Uncertainties

Each result is the mean over the independently pretrained models of a configuration: three for each level of the hierarchy, two for the models with the mass regression and for each partition with controlled flavor boundaries, one for each random partition and each kinematic-input model. For three runs we give the standard deviation, rounded following the Particle Data Group [50] and never to more than four significant figures; for two runs both values. A configuration is compared with another by the ratio of $1-\text{AUC}$ between models with the same seed, which share their training jets and initial weights, with the mean and standard deviation over seeds; a random partition is compared with the 188-class model of its seed. We call two configurations different when the ratio differs from one by more than twice its standard deviation and by more than its test-sample uncertainty, which a bootstrap over the test jets shared by all models gives (Appendix G); results from two runs are reported as observations. Each pretrained model is probed or fine-tuned once, so the spread of the linear probes measures the variation

Table 2: Linear probes on the frozen class token at the largest training set of each task: $(1 - \mathrm{AUC}) \times 10^3$, lower is better. JetClass-II: the macro AUC over its 188 classes, and over its resonant and QCD classes separately; JetClass: the macro AUC; top and quark/gluon (q/g) tagging: the AUC, q/g also on jets generated with Herwig. Mean ± standard deviation over three pretraining runs; for configurations with two runs both values are given, and a random partition has a single run. *: a probe that did not converge.

pretraining	JetClass-II 1,000,000 jets	resonant 1,000,000 jets	QCD 1,000,000 jets	JetClass 1,000,000 jets	top 1,200,000 jets	q/g 1,600,000 jets	q/g, Herwig 1,600,000 jets
188 classes	24.01 ± 0.04	19.05 ± 0.03	53.5 ± 0.4	31.9 ± 0.6	59 ± 8	122 ± 6	196.2 ± 1.7
162 classes	25.19 ± 0.31	19.06 ± 0.11	61.7 ± 2.0	31.76 ± 0.21	52 ± 6	131 ± 9	200 ± 6
43 classes	28.1 ± 0.6	21.69 ± 0.08	66 ± 4	32.0 ± 0.4	62 ± 9	129 ± 6	199.6 ± 2.7
17 classes	36.9 ± 0.6	27.58 ± 0.35	93 ± 4	35.0 ± 0.8	58 ± 4	134 ± 11	201 ± 5
162 classes, mass regression	26.51, 26.03	19.63, 19.29	67.6, 66.3	31.6, 30.4	50.9, 47.1	123.6, 125.2	194.5, 199.2
17 classes, mass regression	37.65, 36.80	28.27, 27.78	93.6, 90.6	34.46, 34.89	41.8, 51.8	126.5, 128.1	199.6, 199.2
random partition 1, seed 1	28.1	21.0	70.5	36.5	70.4	125	198
random partition 2, seed 2	28.1	20.9	70.7	36.0	57.4	131	201
random partition 3, seed 1	27.6	20.7	68.8	36.2	61.3	123	201
random partition 4, seed 2	27.7	20.7	69.1	36.5	81.6	140	204
random partition 5, seed 2	28.0	21.0	69.5	36.1	53.0	138	206
random partition, no flavor boundary	34.3, 35.5	25.89, 26.54	84.7, 89.1	36.72, 37.71	58.0, 54.7	123.8, 125.2	195.9, 199.6
random partition, *b*-quark boundary	30.20, 29.95	22.70, 22.58	74.9, 73.9	33.99, 33.50	51.6, 49.5	132.9, 125.6	202.1, 198.2
random partition, random flavor boundary	30.18, 30.15	22.27, 22.25	77.32, 77.31	34.61, 35.51	59.7, 67.6	126.4, 133.4	196.2, 203.3
randomly initialized network	194.2 ± 1.8	180.9 ± 1.7	273 ± 4	172.2 ± 1.0	61.1 ± 0.5	138.8 ± 1.6	203.5 ± 0.9
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]

between pretraining runs, and after fine-tuning it also contains that of one fine-tuning run; separating the two would need several fine-tuning runs per pretraining run, as in Ref. [57, App. D]. The runs trained on more than one GPU type; the three 188-class runs, two on one type and one on another, agree on JetClass-II to 0.2%, two orders of magnitude below the differences between label sets, which bounds the effect of the GPU type.

4 Results

4.1 Linear probes

Table 2 gives the linear-probe performance on the largest training set of each task for every pretrained configuration and for the randomly initialized network, Table 3 the nonlinear probes, and Fig. 3 the dependence on the training-set size. On JetClass-II, $1 - \mathrm{AUC}$ grows as the label set becomes coarser. Relative to the 188-class models, it is 1.049 ± 0.014 at 162 classes, 1.171 ± 0.024 at 43 classes and 1.538 ± 0.025 at 17 classes, and the label sets keep this order at every training-set size (Fig. 3). The cost of merging the QCD classes lies in the QCD classes alone. On the resonant classes, the ratio of $1 - \mathrm{AUC}$ between the 162- and 188-class models is 1.000 ± 0.006, and on the QCD classes it is 1.153 ± 0.035; the nonlinear probes give 0.998 ± 0.006 and 1.11 ± 0.05. Merging quark flavors raises $1 - \mathrm{AUC}$ on the resonant

Table 3: Nonlinear probes (a multilayer perceptron with a single hidden layer, trained until its validation AUC stops improving, with three seeds whose scores are averaged) at the largest training set of each task: $(1 - \text{AUC}) \times 10^3$, lower is better, with the macro AUC on JetClass-II and JetClass. Upper part: the tasks of Table 2. Lower part: the binary probes between pairs of JetClass-II classes of Table 4, on the same jets; which label sets merge each pair are marked there. The three 188-class runs agree on $X \to bc$ to a few 10^{-6} in AUC although their scores differ; the value was re-derived from their features. Mean $\pm$ standard deviation over three pretraining runs; for configurations with two runs both values are given, and a random partition has a single run.

pretraining	JetClass-II	resonant	QCD	JetClass	top	q/g	q/g, Herwig
188 classes	23.27 ± 0.16	18.59 ± 0.03	51.2 ± 1.2	21.4 ± 0.4	28.7 ± 3.4	106 ± 4	192.3 ± 0.6
162 classes	24.05 ± 0.35	18.56 ± 0.10	56.8 ± 2.3	21.43 ± 0.18	29 ± 4	111 ± 5	194.0 ± 1.4
43 classes	26.4 ± 0.5	20.68 ± 0.07	61 ± 4	21.78 ± 0.29	31.2 ± 2.9	109.6 ± 3.1	192.9 ± 0.7
17 classes	[no run yet]	[no run yet]	[no run yet]	[no run yet]	31.2 ± 1.8	113 ± 6	193.6 ± 2.5
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 3	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 4	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 5	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, no flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, b-quark boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, random flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
randomly initialized network	[no run yet]	[no run yet]	[no run yet]	[no run yet]	33.3 ± 0.6	115.6 ± 0.2	192.6 ± 0.7
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]

pretraining	$X \to bc$ vs bq, cs	$X \to b\bar{b}$ vs $c\bar{c}$	$X \to YY \to bbqq$ vs $ccqq$	QCD bb vs cc	QCD with b vs light	QCD with c, no b, vs light
188 classes	$57.70 \pm <0.01$	2.68 ± 0.11	6.9 ± 0.5	15.4 ± 1.1	13.4 ± 0.7	69.1 ± 0.9
162 classes	57.91 ± 0.21	2.68 ± 0.13	6.70 ± 0.21	16.5 ± 0.8	14.3 ± 0.9	71.9 ± 0.6
43 classes	65.5 ± 1.0	2.98 ± 0.12	7.07 ± 0.04	17.4 ± 0.4	14.1 ± 1.1	73.07 ± 0.21
17 classes	101.8 ± 2.5	8.75 ± 0.34	23.4 ± 2.7	41.7 ± 1.7	17.6 ± 1.4	90.1 ± 0.9
162 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
17 classes, mass regression	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 1	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 2	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 3	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 4	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition 5	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, no flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, b-quark boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
random partition, random flavor boundary	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]
randomly initialized network	408 ± 7	278 ± 16	306 ± 10	388 ± 24	219.2 ± 1.4	331 ± 8
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]	[no run yet]

classes by factors of 1.138 ± 0.004 and 1.448 ± 0.019 at 43 and 17 classes. On the QCD classes, which the 162-class label set already merges, the ratio to the 162-class models is 1.08 ± 0.10 at 43 classes, no cost beyond that of the QCD merge, and 1.50 ± 0.11 at 17 classes. Every pretrained model is far better than the randomly initialized network: the 188-class models reach 0.1236 ± 0.0012 of its $1 - \text{AUC}$.

The random partitions show that on JetClass-II the probes depend on the flavor boundaries of the labels more than on the number of classes (Fig. 4). Every partition in this comparison has 17 classes with the group shares of the 17-class label set. Relative to the 188-class models, $1 - \text{AUC}$ is 1.161 ± 0.009 for the random partitions, which have many flavor bound-

aries, the same as for the 43-class models, which have more classes: paired by seed, the ratio between them is 1.000 ± 0.016. On the resonant classes alone, the random partitions are better than the 43-class models, with a ratio of 0.963 ± 0.007. The ratio to the 188-class models is 1.432 and 1.477 for the random partition with no flavor boundary, below the 1.538 ± 0.025 of the 17-class models in both of its runs, and for the partitions with a b-quark boundary (1.260 and 1.245) and with a random flavor boundary (1.259 and 1.254) it lies between that of the random partitions and that of the partition with no flavor boundary.

On JetClass, which is simulated differently and has different classes, merging quark flavors costs less. The 162- and 43-class models match the 188-class models, with ratios of $1 - \text{AUC}$ of 0.997 ± 0.023 and 1.004 ± 0.031, and only the 17-class models are worse, by a factor of 1.10 ± 0.04. The random partitions, despite their many flavor boundaries, are no better than the 17-class models: their ratio to the 188-class models is 1.152 ± 0.011, and to the 17-class model of the same seed 1.032 ± 0.028. The random partition with no flavor boundary gives 1.155 and 1.206. It differs from the 17-class label set only in how it groups the decays, at random rather than by topology, and its ratio to the 17-class models is 1.074 and 1.053, higher in both runs. On JetClass, the flavor boundaries of the random partitions therefore bring no measurable gain.

On top tagging, no label set gives frozen features that are better than those of the randomly initialized network by more than our threshold for a difference. On the largest training set, the ratio of $1 - \text{AUC}$ to the randomly initialized network is 0.97 ± 0.14 for the 188-class models and between 0.85 ± 0.10 and 1.02 ± 0.14 for the other levels; the nonlinear probes give 0.86 ± 0.11 for the 188-class models. The pretrained models do not pull ahead on smaller training sets either (Fig. 3), and the spread between pretraining runs is larger than that between random initializations. Each of the 162-class runs lies below the randomly initialized network, but by less than our threshold for a difference. On quark/gluon tagging, the 188-class models reduce $1 - \text{AUC}$ to 0.88 ± 0.05 of its value for the randomly initialized network, and to 0.92 ± 0.04 with the nonlinear probes; for the coarser label sets the reduction is below our threshold for a difference. Keeping gluons apart from light quarks in the labels gives no consistent advantage: relative to the 188-class models, $1 - \text{AUC}$ is 1.07 ± 0.11 at 162 classes, which also keep them apart, and 1.06 ± 0.09 and 1.10 ± 0.08 at 43 and 17 classes, which merge them. On the Herwig sample, the gain from pretraining vanishes once the probe is nonlinear: $1-\text{AUC}$ of the 188-class models is 0.964 ± 0.012 of that of the randomly initialized network with the linear probes and 0.9985 ± 0.0032

Table 4: Binary linear probes between JetClass-II classes on the largest training subset: $(1-\mathrm{AUC})\times 10^3$, lower is better. †: the label set merges the two sides into one class; ‡: it merges the first side with part of the second. Mean $\pm$ standard deviation over three pretraining runs; for configurations with two runs both values are given, and a random partition has a single run. The b-quark boundary and the random flavor boundary lie only in the four-quark decays $X \to YY \to QQQQ$; every label set but the 188-class one merges all QCD classes.

pretraining	$X \to bc$ vs bq, cs	$X \to b\bar{b}$ vs $c\bar{c}$	$X \to YY \to bbqq$ vs $ccqq$	QCD bb vs cc	QCD with b vs light	QCD with c, no b, vs light
188 classes	59.46 ± 0.17	2.59 ± 0.07	6.30 ± 0.13	14.88 ± 0.23	11.72 ± 0.20	68.71 ± 0.28
162 classes	59.49 ± 0.15	2.634 ± 0.018	6.44 ± 0.27	$17.1 \pm 0.6^\dagger$	$12.67 \pm 0.18^\dagger$	$73.05 \pm 0.31^\dagger$
43 classes	$68.7 \pm 1.5^\ddagger$	2.810 ± 0.018	6.90 ± 0.09	$17.69 \pm 0.04^\dagger$	$12.66 \pm 0.17^\dagger$	$73.60 \pm 0.30^\dagger$
17 classes	$118 \pm 4^\dagger$	$10.5 \pm 0.7^\dagger$	$29.7 \pm 3.3^\dagger$	$47.9 \pm 1.6^\dagger$	$18.5 \pm 1.1^\dagger$	$96.9 \pm 1.0^\dagger$
random partition 1, seed 1	$69.5^\dagger$	$3.93^\dagger$	$9.18^\dagger$	$22.4^\dagger$	$13.7^\dagger$	$79.5^\dagger$
random partition 2, seed 2	$71.3^\ddagger$	4.22	9.10	$23.9^\dagger$	$14.6^\dagger$	$80.6^\dagger$
random partition 3, seed 1	63.8	$4.12^\dagger$	$8.81^\dagger$	$22.0^\dagger$	$14.4^\dagger$	$78.3^\ddagger$
random partition 4, seed 2	64.2	3.48	8.27	$21.4^\dagger$	$14.1^\dagger$	$77.4^\ddagger$
random partition 5, seed 2	$73.6^\dagger$	$4.53^\dagger$	9.58	$23.9^\dagger$	$13.9^\dagger$	$79.0^\dagger$
random partition, no flavor boundary	$120, 132^\dagger$	$16, 32^\dagger$	$41, 62^\dagger$	$60, 77^\dagger$	$18.78, 17.82^\dagger$	$92.4, 88.7^\dagger$
random partition, b-quark boundary	$91.69, 92.52^\dagger$	$4.79, 4.49^\dagger$	9.08, 8.98	$22.78, 22.46^\dagger$	$13.99, 14.33^\dagger$	$85.12, 84.78^\dagger$
random partition, random flavor boundary	$92.15, 91.55^\dagger$	$6.6, 8.4^\dagger$	$14.20, 14.70^\dagger$	$30.4, 32.3^\dagger$	$15.03, 15.54^\dagger$	$80.68, 80.67^\dagger$

with the nonlinear probes. On both benchmarks, the inputs lack features that every pretraining jet has (Section 3.3); Section 4.3 measures what this costs on JetClass-II, and Section 4.5 shows that fine-tuning recovers a gain from pretraining.

4.2 Separation of merged classes

Table 4 and Fig. 5 give the binary probes between pairs of JetClass-II classes.

Pretraining on the 17-class label set, which merges every pair in the table, increases $1-\mathrm{AUC}$ relative to the 188-class models by factors of 1.98 ± 0.06, 4.06 ± 0.35 and 4.7 ± 0.5 for the resonant pairs in the order of the table, and by 3.22 ± 0.11, 1.58 ± 0.11 and 1.411 ± 0.019 for the QCD pairs. The 43-class label set merges $X \to bc$ with $X \to bq$ but keeps the other resonant pairs apart; pretraining on it increases $1-\mathrm{AUC}$ most for the pair it partly merges, by a factor of 1.156 ± 0.028, and by 1.085 ± 0.027 and 1.095 ± 0.032 for the pairs it keeps apart. The 162-class label set keeps every resonant pair apart and separates them as the 188-class models do (1.000 ± 0.005, 1.017 ± 0.029 and 1.02 ± 0.06). It merges the QCD classes, and the linear probes separate the QCD pairs slightly worse, by factors of 1.15 ± 0.05, 1.081 ± 0.029 and 1.063 ± 0.007. The nonlinear probes change in the same direction, by 1.08 ± 0.09, 1.06 ± 0.06 and 1.041 ± 0.021, but by less than our threshold for a difference. The nonlinear probes reproduce the effects of the 17-class label set on every pair but QCD jets with a b quark against light QCD jets, where the factor, 1.32 ± 0.18, is in the same direction but below

the threshold (Table 3).

For the five random partitions, which have the group shares of the 17-class label set but random groups with many flavor boundaries, merging a pair changes its separation by no more than the spread between partitions, except, in all of the partitions concerned, for $X \to bc$. In the units of Table 4, $1 - \text{AUC}$ for $X \to b\bar{b}$ against $X \to c\bar{c}$ is 4.19 ± 0.31 averaged over the partitions that merge the pair and 4.22 and 3.48 over those that keep it apart, and for $X \to YY \to bbqq$ against $X \to YY \to ccqq$ it is 9.18 and 8.81 and 9.0 ± 0.7. Only $X \to bc$ is separated worse in both partitions that merge it with both $X \to bq$ and $X \to cs$ than in both that keep it apart (69.5 and 73.6 against 63.84 and 64.15), an observation from 2 partitions on each side. These averages are not paired by seed: for $X \to YY \to bbqq$ against $X \to YY \to ccqq$, the partitions that merge the pair share a seed and those that keep it apart share another, so for this pair the effect of merging cannot be separated from that of the seed. After pretraining on any of these partitions, the probes separate every pair better than after pretraining on the 17-class label set, including the QCD pairs, which every random partition merges into a single QCD class. This suggests that the 17-class models separate these pairs poorly not because the pairs are merged but because the 17-class label set has no flavor boundary. The random partitions have flavor boundaries elsewhere, and a network trained on them also separates b from c quarks in decays that share a label.

The random partitions with controlled flavor boundaries test this interpretation. After pretraining on the random partition with no flavor boundary, the linear probes separate $X \to bc$ from $X \to bq$ and $X \to cs$ about as poorly as after pretraining on the 17-class label set, and they separate the other resonant pairs and the QCD bb and cc classes worse than the 17-class models do, in both of its runs. The random partition with a b-quark boundary adds a single flavor boundary, in the four-quark decays, together with a compensating move of other decays, and it still merges $X \to b\bar{b}$ with $X \to c\bar{c}$. Pretraining on it reduces $1 - \text{AUC}$ for this two-prong pair to 0.31 and 0.14 of its value for the partition with no flavor boundary, to values at or just above the upper edge of the range of the five random partitions. The random partition with a random flavor boundary makes the same compensating move but keeps apart no four-quark decays that differ only by exchanging b and c quarks. It reduces $1 - \text{AUC}$ for the same pair to 0.43 and 0.26 of its value for the partition with no flavor boundary, so in both runs a boundary between the four-quark decays with and without a b quark gives the larger reduction, an observation from two runs; the ratio between these partitions is 1.38 and 1.87. Part of both reductions may come from the

Table 5: Linear probes on the 188-class JetClass-II task with 10^6 training jets: $(1 - \text{AUC}) \times 10^3$ on the features as computed, and with the input features that the top-tagging and the quark/gluon data lack set to zero, as in those data. Mean ± standard deviation over the three pretraining runs of each level of the hierarchy and the three randomly initialized networks.

pretraining	all features	top-tagging features	quark/gluon features
188 classes	24.01 ± 0.04	**[no run yet]**	**[no run yet]**
162 classes	25.19 ± 0.31	202.4, 197.8 **[2 of 3]**	107 **[1 of 3]**
43 classes	28.1 ± 0.6	**[no run yet]**	**[no run yet]**
17 classes	36.9 ± 0.6	**[no run yet]**	**[no run yet]**
randomly initialized network	194.2 ± 1.8	246 ± 6	194.5 ± 3.1

compensating move, which adds no flavor boundary. For $X \to bc$ against $X \to bq$ and $X \to cs$, both partitions reduce $1-\text{AUC}$ less, to 0.764 and 0.702 and 0.768 and 0.695 of its value for the partition with no flavor boundary (Table 4).

4.3 Missing input features

[the masking control: JetClass-II probes with the features that the top-tagging and quark/gluon data lack set to zero, for every level of the hierarchy and the randomly initialized network]

4.4 Auxiliary mass regression

Table 6 gives the mass resolution. The uncorrected soft-drop mass has $\sigma_{\text{eff}} = 0.253$ and a median response of 0.929. The mass output of the models pretrained with the mass regression reduces σ_{eff} to 0.5811 and 0.5779 of this value at 162 classes and 0.5728 and 0.5768 at 17 classes, and brings the median response to 0.995 and 1.003 and 1.005 and 1.005. Frozen features carry less of this information. For the 188-class models, the nonlinear regression on the frozen features reaches 0.8818 ± 0.0032 of the σ_{eff} of a prediction by the class median alone, and for the randomly initialized network 0.973 ± 0.004. Without the mass regression, the 17-class models resolve the mass slightly better than the 188-class models, with a ratio of σ_{eff} of 0.955 ± 0.012. With it, σ_{eff} of the regression on the frozen features falls to 0.711 and 0.727 of its value without it at 162 classes and 0.757 and 0.740 at 17 classes. The gain is similar at both label sets, although the mass term makes up a larger share of the training loss at 17 classes (Section 3.2).

Table 6: Mass resolution σ_{eff} of $\ln(m_{\text{reg}}/m_{\text{SD}}^{\text{gen}})$ on the JetClass-II test jets with a generator-level match, on all, resonant and QCD jets, and the median $m_{\text{reg}}/m_{\text{SD}}^{\text{gen}}$; lower σ_{eff} is better. The mass node of the models pretrained with the mass regression is scored on the target itself. The multilayer perceptron (MLP) on frozen features, averaged over three seeds, regresses the target minus the median of the jet's true class and is scored with that median added back, so its rows are read against the class median alone, not against the uncorrected mass or the mass node. Mean ± standard deviation over three pretraining runs; the models with the mass regression have two runs, and both values are given.

pretraining	σ_{eff}	resonant	QCD	$m_{\text{reg}}/m_{\text{SD}}^{\text{gen}}$
uncorrected soft-drop mass	0.253	0.221	0.478	0.929
mass node				
162 classes, mass regression	0.1473, 0.1464	0.1323, 0.1315	0.2257, 0.2247	0.995, 1.003
17 classes, mass regression	0.1451, 0.1462	0.1304, 0.1312	0.2228, 0.2233	1.005, 1.005
MLP on frozen features, with the true class's median added back				
class median alone	0.251	–	–	–
188 classes	0.2212 ± 0.0008	0.1993 ± 0.0011	0.3394 ± 0.0030	–
162 classes	0.2203 ± 0.0024	0.1986 ± 0.0018	0.336 ± 0.004	–
43 classes	0.2161 ± 0.0025	0.1944 ± 0.0019	0.333 ± 0.006	–
17 classes	0.2113 ± 0.0021	0.1901 ± 0.0026	0.322 ± 0.004	–
162 classes, mass regression	0.1585, 0.1585	0.1418, 0.1420	0.2454, 0.2442	–
17 classes, mass regression	0.1581, 0.1575	0.1408, 0.1412	0.2456, 0.2430	–
randomly initialized network	0.2440 ± 0.0009	0.2203 ± 0.0011	0.3855 ± 0.0023	–

The mass regression changes the linear probes on JetClass-II and JetClass little: $1 - \text{AUC}$ changes by factors of 1.041 and 1.032 and 1.034 and 1.002 on JetClass-II at 162 and 17 classes, and by 0.989 and 0.958 and 1.008 and 0.974 on JetClass. On top tagging, $1 - \text{AUC}$ falls in both runs at both label sets, to 0.908 and 0.863 and 0.72 and 0.84 of its value without the mass regression at 162 and 17 classes. With two runs and the spread of the top-tagging probes between runs, this is an observation rather than an established difference.

Table 7 gives the mass sculpting. A cut on the score of every model reshapes the soft-drop mass distribution of the QCD jets it accepts, and at a QCD efficiency of 10% the label set makes no difference: the ratio of the divergence between the 17- and 188-class models is 1.08 ± 0.11. At 1%, only 4,103 QCD jets pass, and a random selection of as many jets already gives a divergence of 1.01 in the units of the table, against 0.101 at 10%. Above this level, the 17-class models sculpt more than the 188-class models, by a factor of 1.64 ± 0.11 before the level is subtracted. The mass regression reduces

Table 7: Mass sculpting: Jensen–Shannon divergence $\times 10^3$ between the m_{SD} distributions (24 bins of 20 GeV from 20 to 500 GeV) of all QCD test jets and of those passing a cut on $1 - p_{\mathrm{QCD}}$ at the QCD efficiency given; lower is less sculpting. The last row is the divergence expected for a random selection of as many QCD jets, which a cut that does not sculpt the mass still shows because of the finite number of jets passing it. Mean ± standard deviation over three pretraining runs; the models with the mass regression have two runs, and both values are given.

pretraining	$\epsilon_{\mathrm{QCD}} = 0.1$	$\epsilon_{\mathrm{QCD}} = 0.01$
188 classes	2.8 ± 0.4	1.93 ± 0.21
162 classes	2.83 ± 0.10	1.84 ± 0.24
43 classes	3.14 ± 0.30	2.6 ± 0.7
17 classes	2.95 ± 0.11	3.15 ± 0.20
162 classes, mass regression	1.78, 2.15	1.80, 1.95
17 classes, mass regression	2.72, 1.79	1.64, 2.04
random selection of as many QCD jets	0.101	1.01

the sculpting at 10% in both runs at both label sets, by factors of 0.62 and 0.74 at 162 classes and 0.89 and 0.62 at 17 classes. At 1%, it reduces the sculpting of the 17-class models by factors of 0.49 and 0.68, to 1.64 and 2.04, about the level of the 188-class models (1.93 ± 0.21).

4.5 Fine-tuning

Table 8 and Fig. 6 give the fine-tuned models against the training-set size, and Table 9 gives their background rejection on the full training sets. On the full training sets, the network trained from scratch reproduces ParT trained from scratch on quark/gluon tagging, with an AUC of 0.9121 ± 0.0002 against 0.9121 [54, Table 6], and is below it on top tagging, with 0.9852 ± 0.0001 against 0.9858 [54, Table 5] and a lower background rejection, given below. Fine-tuning the 188-class models gives 0.9148 ± 0.0001 and 0.9861 ± 0.0001, against 0.9151 and 0.9877 for ParT pretrained on JetClass. On top tagging the gain from pretraining is smaller than ParT's: the background rejection at 30% signal efficiency rises from $1{,}412 \pm 20$ to $1{,}800 \pm 60$, against 1602 to 2766 for ParT. ParT pretrained a separate model on the kinematic features alone for top tagging, whereas our models were pretrained on all features and receive the missing ones as zeros (Section 3.3), and they were pretrained on JetClass-II, with pileup, rather than on JetClass.

On quark/gluon tagging, fine-tuning the 188-class models beats training from scratch at every training-set size: $1-\mathrm{AUC}$ is 0.892 ± 0.009 of that of the network trained from scratch at 10^3 jets, 0.965 ± 0.013 at 10^4 and 0.9693 ± 0.0021 on the full training set; at 10^4 jets the 17-class models, with 0.985 ± 0.018, do not pass our threshold. On top tagging, the ratio is 0.73 ± 0.07 at 10^3 jets, 0.899 ± 0.010 at 10^5 and 0.944 ± 0.006 on the full training set, and 1.09 ± 0.08 at 10^4 jets, where pretraining brings no gain within the spread of the network trained from scratch. At 10^3 and 10^4 jets, every model is trained for the same small number of steps, the pretrained weights at a learning rate of 10^{-4} and the network trained from scratch at 10^{-3} (Appendix C). Fine-tuned for longer at 10^4 jets, the 188-class models reach **[recipe check]** in the units of Table 8 and the network trained from scratch **[recipe check]**.

After fine-tuning, the label set matters little. The 17-class models have a higher $1-\mathrm{AUC}$ than the 188-class models at every size of both tasks, by a ratio of at most 1.19 ± 0.07 (top tagging, 10^4 jets); on the full training sets it is 1.012 ± 0.002 on top and 1.005 ± 0.001 on quark/gluon tagging, and at 10^3 jets the difference is below our threshold. The 162- and 43-class models match the 188-class models on the full training sets, with ratios of 0.9975 ± 0.0007 and 0.9975 ± 0.0014 on top and 1.000 ± 0.003 and 1.003 ± 0.003 on quark/gluon tagging. The random partitions behave like the 17-class models: on the full training sets their ratio to the 188-class models is 1.009 ± 0.005 on top and 1.004 ± 0.001 on quark/gluon tagging, and on the small top-tagging sets it is 1.26 ± 0.11 at 10^3 and 1.33 ± 0.14 at 10^4 jets, where their ratio to the 17-class model of the same seed, 1.11 ± 0.12 and 1.16 ± 0.12, is below our threshold. On the Herwig sample, the models fine-tuned on Pythia jets keep little of the gain from pretraining: on the full training set, the ratio to the network trained from scratch is 0.991 ± 0.012 for the 188-class models. The lower $1-\mathrm{AUC}$ of the frozen top-tagging features with the auxiliary mass regression, observed in both runs (Section 4.4), does not survive fine-tuning: on the full training set, $1-\mathrm{AUC}$ with the mass regression is 1.008 and 1.009 and 1.008 and 1.006 of its value without it at 162 and 17 classes.

[fine-tuning on JetClass-II at its 188 classes and on JetClass]

4.6 Resonance tagging

The transfer classifier is trained on **[samples not in]** jets of each class for $X \to b\bar{b}$ and **[samples not in]** for $X \to bs$, and every model is evaluated on the same **[samples not in]** and **[samples not in]** signal jets and **[samples**

not in] QCD jets in the window of **[samples not in]**–**[samples not in]** GeV (Table 12, Fig. 7). **[the result for each label set once the samples are in]**

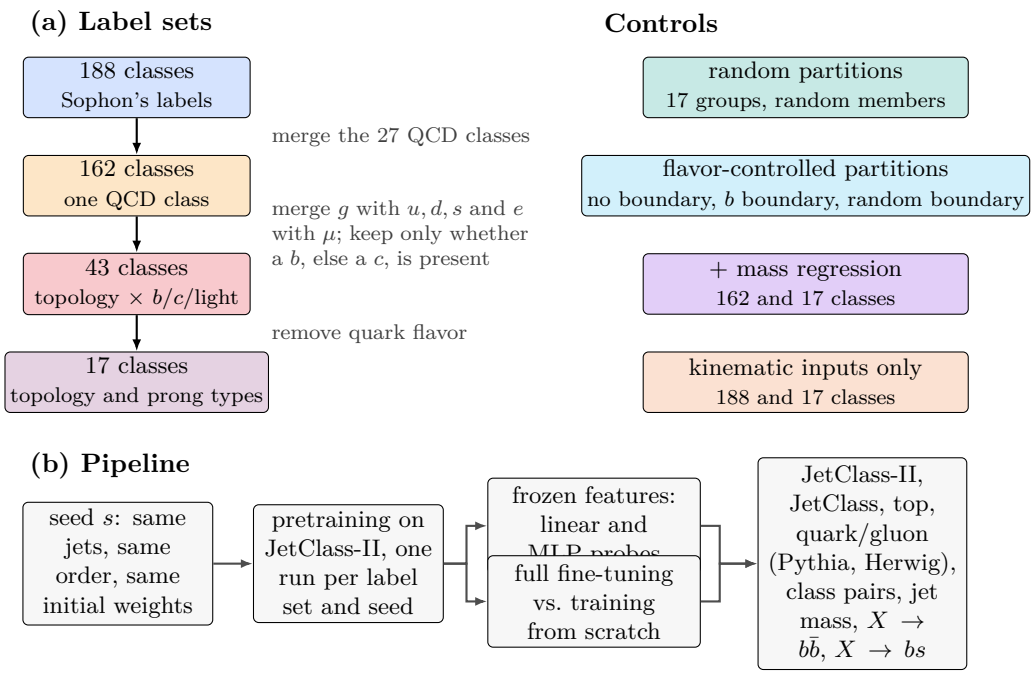


Figure 1: Design of the study. (a) Left: four levels of one class hierarchy of the 188 JetClass-II classes and what each merge removes. Every class holds only QCD jets, only decays into quarks and gluons, or only decays with leptons. Right: the controls. The random partitions keep the group sizes of the 17-class level but assign the decays to the groups at random; they have many flavor boundaries and no topological order. The flavor-controlled partitions fix where the flavor boundaries lie. Two further variants add a mass-regression output or remove the particle-type and track-displacement inputs. Figure 2 shows the classes of selected decays in each label set. (b) Models with the same seed see the same training jets in the same order from the same initial weights, so two models of one seed differ only in their labels and output layer. Every model is evaluated with probes on its frozen features and with full fine-tuning on the same tasks.

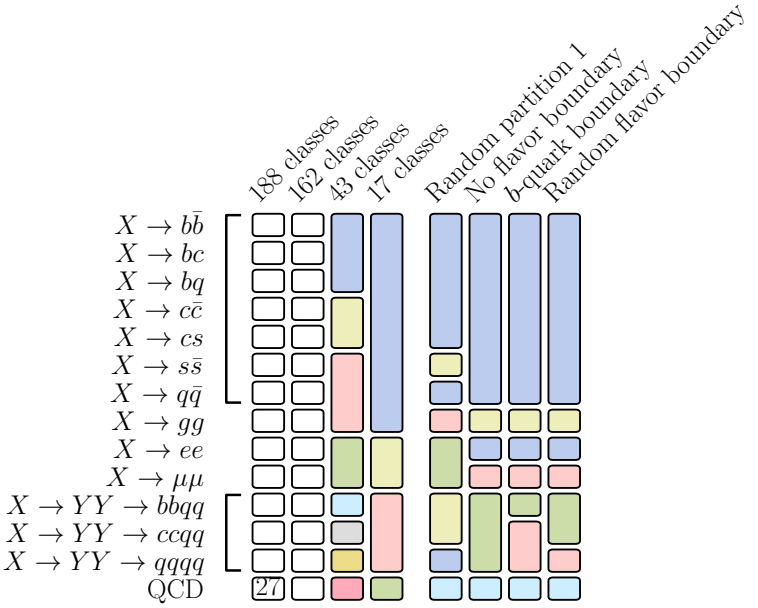


Figure 2: Classes of selected decays in the label sets of the class hierarchy (left) and in random partitions (right): random partition 1 and the partitions with no flavor boundary, with a b-quark boundary and with a random flavor boundary. In each colored column, decays of the same color share a class, and adjacent decays of the same class form a single box; colors are compared only within a column. In the 188- and 162-class label sets (white boxes), every resonant decay is its own class, and the 188-class label set divides the QCD jets into 27 classes by their quark content. Brackets mark decays that differ only in quark flavor, so a change of color inside a bracket is a flavor boundary.

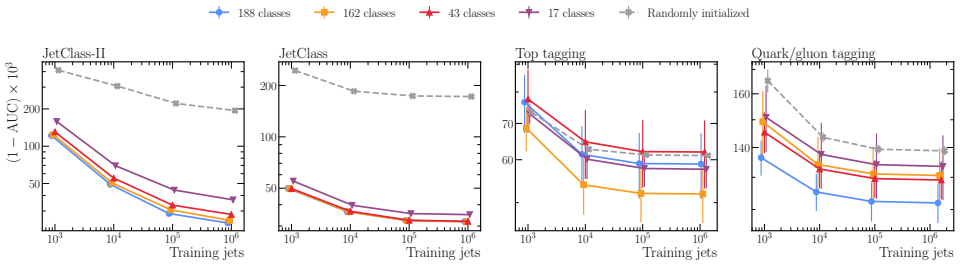


Figure 3: Linear-probe $1 - \mathrm{AUC}$ against the number of training jets for each level of the hierarchy and for the randomly initialized network, scaled as in Table 16: from left, JetClass-II (macro AUC over all classes), JetClass (macro AUC), top tagging and quark/gluon tagging. Both axes are logarithmic. Points are the mean over the pretraining runs and error bars the standard deviation; the series are offset slightly along the horizontal axis so that coinciding points remain visible. At 10^3 and 10^4 training jets, the JetClass-II macro AUC includes classes absent from the training subset, which score as a random guess for every model (Appendix C); Table 16 also gives the average over the classes present, in which the label sets keep the same order.

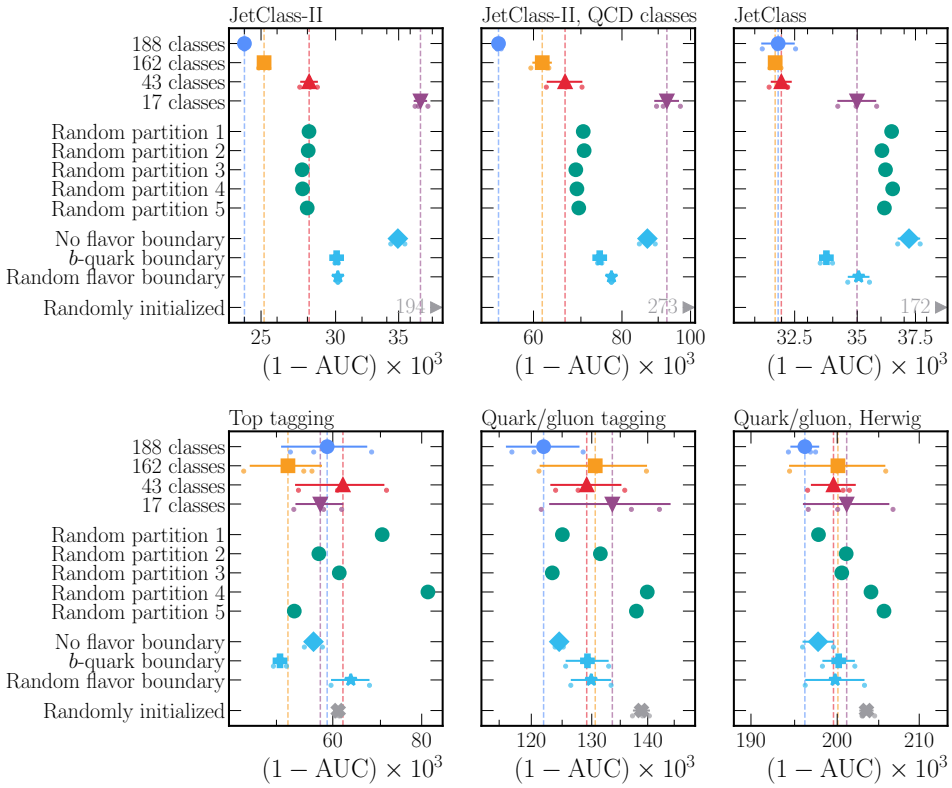


Figure 4: Linear-probe $1 - \mathrm{AUC}$ on the largest training set of each task for every pretraining configuration of Table 2 except those with the auxiliary mass regression, scaled as in that table: the JetClass-II macro AUC over all classes and over its QCD classes, the JetClass macro AUC, and the AUC of top tagging and of quark/gluon tagging on Pythia and on Herwig jets. Markers are the mean over the pretraining runs of a configuration, and error bars the standard deviation over three runs or the range of two; small dots just below each marker show the individual runs, and each random partition has a single run. Dashed lines mark the mean of each level of the hierarchy. Horizontal axes are logarithmic, so equal ratios appear as equal distances. Where the randomly initialized network lies beyond the axis, an arrow at the edge of the panel gives its value.

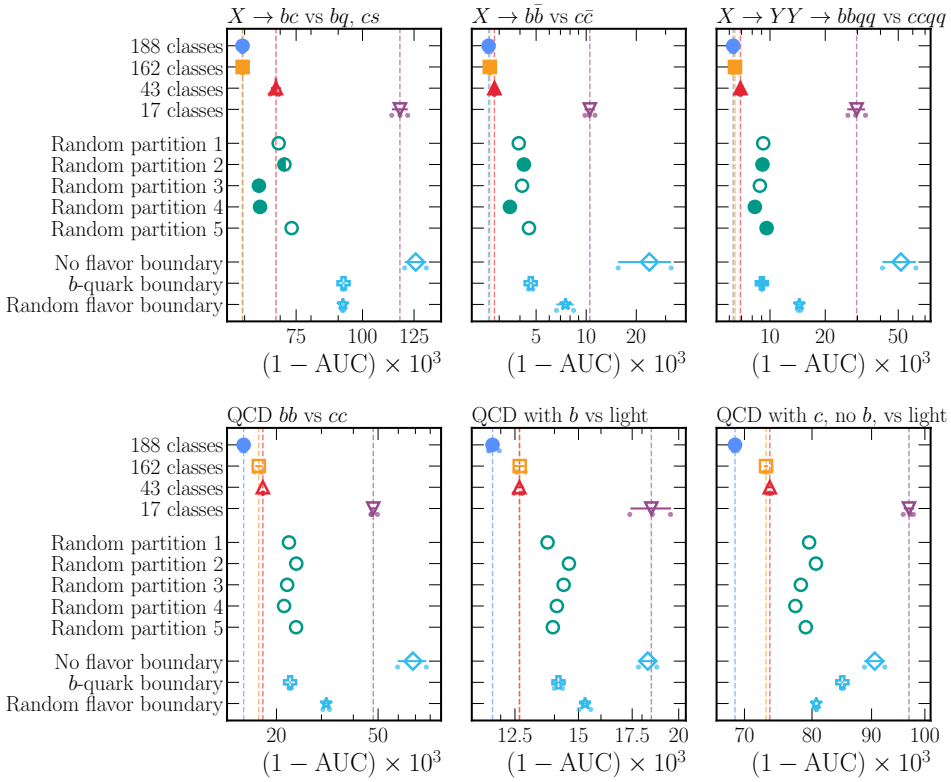


Figure 5: Binary linear-probe $1 - \text{AUC}$ between pairs of JetClass-II classes on the largest training set, scaled as in Table 4, for every pretraining configuration of that table. Open markers show label sets that merge the pair, and partly filled markers label sets that merge part of it; markers, dots, error bars, lines and axes are otherwise as in Fig. 4.

Table 8: Full fine-tuning against the training-set size: $(1 - \text{AUC}) \times 10^3$, lower is better, with the macro AUC (one-versus-rest) on JetClass-II, over all and over its QCD classes, and on JetClass, and the AUC on top and quark/gluon (q/g) tagging, q/g also on jets generated with Herwig. Mean ± standard deviation over three pretraining runs; for configurations with two runs both values are given, and a random partition has a single run. Each pretrained run is fine-tuned once. From scratch: mean ± standard deviation over three seeds, which below the largest training set of JetClass, top and q/g tagging also use different training subsets.

JetClass-II **[not yet read out]**

JetClass-II, QCD classes **[not yet read out]**

JetClass **[not yet read out]**

top	1,000	10,000	100,000	1,200,000
188 classes	47.5 ± 2.3	30.1 ± 1.0	17.17 ± 0.11	13.94 ± 0.07
162 classes	48.3 ± 2.0	32.5 ± 2.1	17.62 ± 0.24	13.90 ± 0.08
43 classes	54 ± 6	34 ± 5	18.06 ± 0.31	13.90 ± 0.05
17 classes	52.3 ± 1.9	35.8 ± 2.7	18.03 ± 0.34	14.10 ± 0.06
162 classes, mass regression	50.4, 45.1	32.6, 30.8	18.32, 17.67	13.99, 13.95
17 classes, mass regression	61.5, 55.7	41.4, 38.9	18.53, 18.98	14.22, 14.13
random partition 1, seed 1	63.7	41.0	18.1	14.1
random partition 2, seed 2	51.8	34.7	18.1	13.9
random partition 3, seed 1	55.3	39.0	18.1	14.0
random partition 4, seed 2	56.7	45.9	18.3	14.1
random partition 5, seed 2	63.7	38.8	18.3	14.0
random partition, no flavor boundary	57.0, 53.0	38.8, 34.5	17.74, 17.78	14.01, 13.91
random partition, b-quark boundary	60, 50	38.6, 29.5	18.05, 17.78	14.01, 13.91
random partition, random flavor boundary	57.3, 64.6	41.98, 42.90	18.74, 18.07	14.17, 13.89
from scratch	65 ± 10	27.7 ± 3.2	19.10 ± 0.31	14.76 ± 0.08

q/g	1,000	10,000	100,000	1,600,000
188 classes	132.0 ± 0.6	114.8 ± 1.4	90.94 ± 0.08	85.18 ± 0.14
162 classes	131.3 ± 1.2	113.7 ± 2.7	91.2 ± 0.5	85.22 ± 0.08
43 classes	133.2 ± 0.9	116.2 ± 1.3	91.8 ± 0.4	85.44 ± 0.32
17 classes	134.8 ± 1.4	117.2 ± 2.0	92.15 ± 0.15	85.60 ± 0.03
162 classes, mass regression	129.8, 131.6	114.6, 114.3	91.64, 91.91	85.31, 85.47
17 classes, mass regression	132.2, 132.7	115.8, 115.8	92.49, 93.28	85.82, 86.00
random partition 1, seed 1	137	114	91.7	85.4
random partition 2, seed 2	144	116	91.6	85.3
random partition 3, seed 1	141	121	92.8	85.5
random partition 4, seed 2	135	118	92.2	85.4
random partition 5, seed 2	138	120	92.8	85.4
random partition, no flavor boundary	127.6, 129.0	113.5, 112.1	91.15, 91.84	85.63, 85.45
random partition, b-quark boundary	132.8, 136.4	117.4, 115.6	91.43, 91.80	85.28, 85.28
random partition, random flavor boundary	137.5, 131.6	117.7, 114.3	91.4, 92.4	85.52, 85.29
from scratch	148.0 ± 2.4	119.0 ± 1.3	98.5 ± 0.8	87.88 ± 0.22

q/g, Herwig	1,000	10,000	100,000	1,600,000
188 classes	206.1 ± 2.3	189.6 ± 1.3	195.0 ± 2.0	192.2 ± 2.3
162 classes	206.8 ± 0.2	189.4 ± 0.8	196.0 ± 1.8	192.6 ± 1.0
43 classes	205.1 ± 2.3	191.1 ± 0.4	196.0 ± 0.3	191 ± 5
17 classes	207.2 ± 1.2	190.1 ± 0.4	191.6 ± 0.7	189.7 ± 1.8
162 classes, mass regression	203.4, 206.0	188.6, 188.8	193.4, 194.6	189.1, 192.6
17 classes, mass regression	204.1, 204.3	189.0, 190.9	191.3, 190.5	189.9, 194.8
random partition 1, seed 1	210	190	196	193
random partition 2, seed 2	222	191	192	192
random partition 3, seed 1	211	196	194	193
random partition 4, seed 2	202	189	196	191

Table 9: Background rejection R_{50} and R_{30} ($1/\epsilon_B$ at 50% and 30% signal efficiency) after full fine-tuning on the whole training set. Mean ± standard deviation over three pretraining runs; for configurations with two runs both values are given, and a random partition has a single run. From scratch: over its seeds. Every value rests on at least 98 background jets passing the cut.

pretraining	top		q/g		q/g, Herwig	
	R_{50}	R_{30}	R_{50}	R_{30}	R_{50}	R_{30}
188 classes	461 ± 4	$1{,}800 \pm 60$	42.56 ± 0.10	107.3 ± 2.2	12.3 ± 0.4	35.2 ± 2.1
162 classes	443 ± 17	$1{,}790 \pm 60$	42.39 ± 0.22	108.0 ± 1.4	12.33 ± 0.28	34.6 ± 2.0
43 classes	450 ± 13	$1{,}780 \pm 150$	42.2 ± 0.8	108.3 ± 1.0	12.5 ± 0.6	36.1 ± 2.7
17 classes	429 ± 10	$1{,}740 \pm 140$	41.97 ± 0.25	106.5 ± 0.8	12.79 ± 0.20	36.9 ± 0.9
162 classes, mass regression	420, 436	1,941, 1,590	42.46, 42.21	107.0, 106.6	12.99, 12.21	37.8, 34.4
17 classes, mass regression	435, 421	1,683, 1,628	41.77, 41.39	104.9, 105.6	12.58, 11.96	37.7, 32.9
random partition 1, seed 1	432	1,660	42.1	107	12.2	33.4
random partition 2, seed 2	430	1,900	42.5	109	12.3	35.2
random partition 3, seed 1	421	1,640	42.1	106	12.2	34.8
random partition 4, seed 2	458	1,820	41.8	107	12.7	37.5
random partition 5, seed 2	431	1,740	42.1	106	12.2	35.3
random partition, no flavor boundary	427, 463	1,726, 1,852	41.67, 41.95	108.9, 109.2	12.07, 12.92	34.1, 39.3
random partition, b-quark boundary	432, 444	1,590, 2,060	42.43, 42.77	106.5, 106.8	12.93, 12.19	38.1, 34.7
random partition, random flavor boundary	394, 447	1,565, 1,655	42.12, 42.09	108.2, 104.4	12.18, 12.76	34.9, 37.4
from scratch	358 ± 11	$1{,}412 \pm 20$	41.47 ± 0.29	104.0 ± 1.0	12.18 ± 0.11	35.5 ± 1.2

Table 10: Top tagging on the full training set, 1.2 million jets: accuracy, AUC and background rejection $1/\epsilon_B$ at 50% and 30% signal efficiency. Published rows from the tables cited; f.t. marks a model pretrained on JetClass and fine-tuned, and the kinematic-input rows were pretrained on the kinematic features only, as ParT-f.t. was. Uncertainties: ours, the standard deviation over pretraining runs, one fine-tuning each (three runs; two runs print both values; one run, the value alone); ParT, the spread over 9 fine-tunings of one pretrained model with different head initializations, quoted for the run of median accuracy; OmniLearn, OmniLearned and L-GATr, the standard deviation of five trainings. The best value of each column is in bold.

	accuracy	AUC	R_{50}	R_{30}
ParticleNet [53]	0.940	0.9858	397 ± 7	1615 ± 93
LorentzNet [30]	0.942	0.9868	498 ± 18	2195 ± 173
ParT [54]	0.940	0.9858	413 ± 16	1602 ± 81
ParticleNet-f.t. [54]	0.942	0.9866	487 ± 9	1771 ± 80
ParT-f.t. [54]	0.944	0.9877	$\mathbf{691 \pm 15}$	2766 ± 130
MIParT-L-f.t. [62]	0.944	0.9878	640 ± 10	2789 ± 133
L-GATr-f.t. [12]	**0.9446**	0.9879	651 ± 11	2894 ± 84
OmniLearn [49]	0.942	0.9872	568 ± 9	2647 ± 192
OmniLearned-s [9]	0.944	0.9875	565 ± 12	2637 ± 128
OmniLearned-m [9]	0.944	**0.9880**	656 ± 12	3208 ± 176
OmniLearned-l [9]	0.944	**0.9880**	688 ± 9	$\mathbf{3486 \pm 157}$
this work (Sophon network, fine-tuned)				
from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$
188 classes	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$
162 classes	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$
43 classes	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$
17 classes	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$
162 classes, mass regression	0.9406, 0.9402	0.9860, 0.9860	420, 436	1,941, 1,590
17 classes, mass regression	0.9397, 0.9399	0.9858, 0.9859	435, 421	1,683, 1,628
random partition, no flavor boundary	0.9402, 0.9401	0.9860, 0.9861	427, 463	1,726, 1,852
random partition, b-quark boundary	0.9401, 0.9405	0.9860, 0.9861	432, 444	1,590, 2,060
random partition, random flavor boundary	0.9397, 0.9406	0.9858, 0.9861	394, 447	1,565, 1,655

Table 11: Quark/gluon tagging on the full training set, 1.6 million jets, with the particle type as five binary flags and the charge (ParT's exp scenario): accuracy, AUC and $1/\epsilon_B$ at 50% and 30% quark efficiency. Published rows from the tables cited, all in the exp scenario; PFN-Ex's uncertainty is half the interquartile range of ten trainings, and the others are as in Table 10, with nine trainings for OmniLearn and OmniLearned. The best value of each column is in bold.

	accuracy	AUC	R_{50}	R_{30}
PFN-Ex [37]	–	0.9005	34.7 ± 0.4	–
ParticleNet [53]	0.840	0.9116	39.8 ± 0.2	98.6 ± 1.3
ParT [54]	0.840	0.9121	41.3 ± 0.3	101.2 ± 1.1
ParticleNet-f.t. [54]	0.839	0.9115	40.1 ± 0.2	100.3 ± 1.0
ParT-f.t. [54]	0.843	0.9151	42.4 ± 0.2	107.9 ± 0.5
OmniLearn [49]	0.844	0.9159	$\mathbf{43.7 \pm 0.3}$	107.7 ± 1.5
OmniLearned-s [9]	0.844	0.9153	42.9 ± 0.2	108.0 ± 0.5
OmniLearned-m [9]	**0.845**	**0.9162**	43.2 ± 0.1	$\mathbf{111.2 \pm 1.5}$
this work (Sophon network, fine-tuned)				
from scratch	0.8397 ± 0.0007	0.9121 ± 0.0002	41.47 ± 0.29	104.0 ± 1.0
188 classes	0.8433 ± 0.0002	0.9148 ± 0.0001	42.56 ± 0.10	107.3 ± 2.2
162 classes	$0.8432 \pm <0.0001$	0.9148 ± 0.0001	42.39 ± 0.22	108.0 ± 1.4
43 classes	0.8430 ± 0.0005	0.9146 ± 0.0003	42.2 ± 0.8	108.3 ± 1.0
17 classes	0.8427 ± 0.0003	$0.9144 \pm <0.0001$	41.97 ± 0.25	106.5 ± 0.8
162 classes, mass regression	0.8425, 0.8427	0.9147, 0.9145	42.46, 42.21	107.0, 106.6
17 classes, mass regression	0.8425, 0.8424	0.9142, 0.9140	41.77, 41.39	104.9, 105.6
random partition, no flavor boundary	0.8423, 0.8434	0.9144, 0.9146	41.67, 41.95	108.9, 109.2
random partition, b-quark boundary	0.8430, 0.8431	0.9147, 0.9147	42.43, 42.77	106.5, 106.8
random partition, random flavor boundary	0.8427, 0.8428	0.9145, 0.9147	42.12, 42.09	108.2, 104.4

Table 12: Resonance tagging in the soft-drop mass window, **[samples not in]**–**[samples not in]** GeV: signal efficiency (%) at the QCD efficiency given, mean ± standard deviation over the pretraining runs. Transfer: Sophon's classifier on the frozen features, its seeds averaged per run (the randomly initialized network: the linear probe). Direct: Sophon's discriminant from the output scores, where $X \to b\bar{b}$ is a class of its own.

[no result yet]

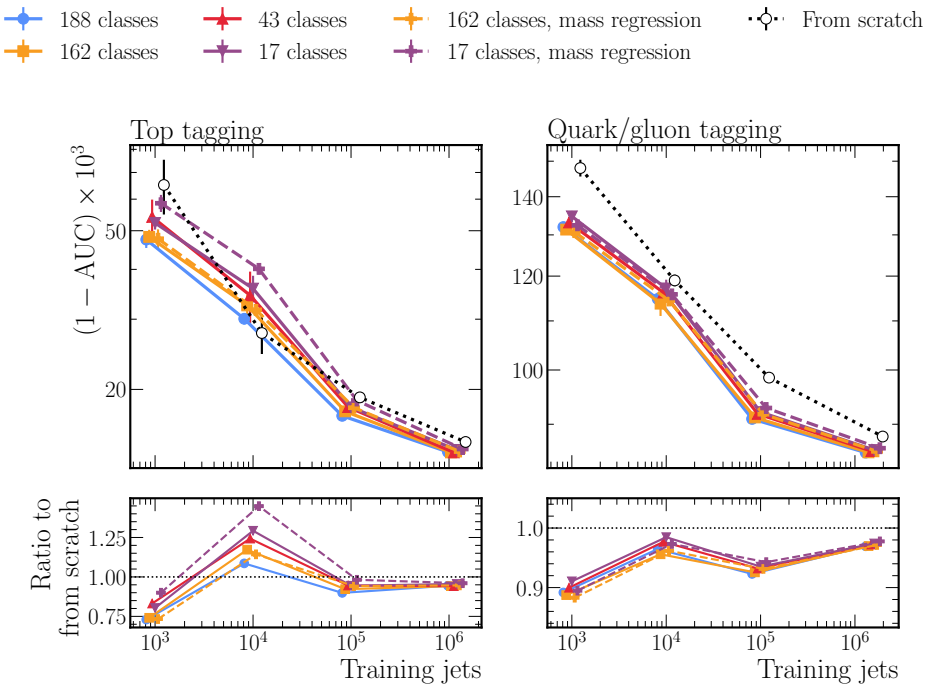


Figure 6: Fine-tuned $1 - \mathrm{AUC}$, scaled as in Table 8, against the number of training jets, a column for each task, for each level of the hierarchy, the models with the auxiliary mass regression and the network trained from scratch (upper row), and the ratio of the mean to that of the network trained from scratch (lower row, without error bars). Horizontal axes are logarithmic, as are the vertical axes of the upper row. Points are the mean over the runs, and error bars the standard deviation over three runs or the range of the two runs with the mass regression; Table 8 gives the values.

[figure xbx: data not in]

Figure 7: Resonance tagging: $\epsilon_S/\sqrt{\epsilon_B}$, proportional to the discovery significance for a small signal, against the QCD efficiency in the mass window. Solid: Sophon's transfer classifier on the frozen features, mean over the pretraining runs with their range shaded; dashed: the direct discriminant of the label sets in which $X \to b\bar{b}$ is a class of its own; dotted (right): the output for the class that contains $X \to bq$, Sophon's control for $X \to bs$.

5 Discussion and conclusion

We retrained the Sophon model on JetClass-II with its 188 classes and with label sets of 162, 43 and 17 classes from a single class hierarchy, with the training examples, their order and the initial weights fixed by the seed, so that models with the same seed differ in their labels and output layer but not in their data or initialization. On JetClass-II, the linear probes degrade steadily as the label set is coarsened, and the random partitions indicate that the representation depends on the flavor boundaries of the labels more than on the number of classes. Merging the QCD classes leaves the probes of the resonant classes unchanged, with a ratio of $1-\mathrm{AUC}$ of 1.000 ± 0.006, and raises $1-\mathrm{AUC}$ on the QCD classes by 1.153 ± 0.035; the nonlinear probes show the same direction. Removing quark flavor from the labels raises $1-\mathrm{AUC}$ by a further factor of 1.47 ± 0.04 at 17 classes, relative to the 162-class label set, and degrades the separation of classes that differ only in flavor, QCD classes included. Random partitions into 17 groups with many flavor boundaries give a ratio of 1.161 ± 0.009, the same as the 43-class label set. A random partition with no flavor boundary gives 1.432 and 1.477, and adding a single flavor boundary to it, together with a compensating move of other decays that may account for part of the change, gives 1.260 and 1.245. On JetClass, the random partitions are no better than the 17-class label set, so there their flavor boundaries bring no measurable gain. On top and quark/gluon tagging, the frozen features of no label set are better than those of a randomly initialized network by more than our threshold for a difference, except for the 188-class models on both quark/gluon samples, and the nonlinear probes reduce that gain further. On the full training sets, fine-tuning does improve on training from scratch on both tasks, reproducing ParT's quark/gluon results, and the label set changes $1-\mathrm{AUC}$ by about a per cent; on the smaller top-tagging sets the 17-class models fall further behind, by up to a factor of 1.19 ± 0.07. An auxiliary mass regression improves the mass resolution of the frozen features by a similar amount at both label sets tested and, in both runs at both label sets, reduces mass sculpting; the lower $1-\mathrm{AUC}$ of its frozen top-tagging features, observed in both runs, does not survive fine-tuning. **[fine-tuning on JetClass-II and JetClass; resonance tagging]**

For the design of pretraining datasets, our results suggest the following. Boundaries between quark flavors should be kept: on JetClass-II, many flavor boundaries bring $1-\mathrm{AUC}$ to 1.161 ± 0.009 of its value for the 188-class models, against 1.538 ± 0.025 for the 17-class label set, and a single flavor boundary, added together with a compensating move of other decays,

recovers at most part of the difference. This advice rests on JetClass-II, whose classes are split by flavor: on JetClass the random partitions are no better than the 17-class label set. Dividing the QCD jets by their quark content improves only the separation of QCD jets by quark content: the QCD macro AUC on JetClass-II with both probes, and the QCD pairs with the linear probes. It leaves the frozen features for the resonant classes, JetClass, top and quark/gluon tagging unchanged within our threshold for a difference, and after fine-tuning on the full training sets it changes $1-\mathrm{AUC}$ on top and quark/gluon tagging by less than a per cent. Whether a dataset needs it therefore depends on whether its downstream tasks separate QCD jets by flavor. Both the 188- and 162-class label sets form every discriminant of Table 1. **[implications of fine-tuning on JetClass-II]**

The study has several limitations. All pretraining and all evaluations except the Herwig quark/gluon sample use a single simulation family, so an effect of the label set that is specific to it cannot be excluded. A single architecture and model size is studied, and whether fine labels help can depend on how strongly a network is overparameterized [52], so a smaller or larger model may respond differently. The label sets are levels of a single hierarchy, and other groupings enter only through the random partitions, with and without controlled flavor boundaries. The models follow Sophon's architecture and training recipe but not its weight initialization, so their absolute performance need not match that of Sophon's published model. Every model is trained on the same number of examples, so the study does not determine whether a finer label set benefits from a longer schedule. With three seeds per level of the hierarchy and fewer for the other configurations, the standard deviations are themselves uncertain, differences between label sets smaller than them are not resolved, and comparisons between random partitions are not paired by seed. The GPU type differs within most pairs of runs that are compared, and its effect on a trained model was not measured. The JetClass-II task uses the 188 pretraining classes themselves, which favors the finest label set; label sets with the same number of classes that keep more of its classes apart, flavor siblings included, share part of this advantage, which the other tasks do not give. On top and quark/gluon tagging, the features that the data lack are set to zero rather than learned without, as in ParT's reduced-input pretraining. Fine-tuning follows a single recipe per dataset, which is not tuned per label set, and its step budget at the smaller training sets is short. **[the $X \to b\bar{b}$ and $X \to bs$ samples, once they are in]**

Data and code availability

Pretrained checkpoints, label maps and evaluation code will be released on Zenodo at submission. Every number in this paper is generated from committed result files by a script that records the hash of each file beside the number.

A Pretraining details

Training examples. Epochs are counted from zero. Training proceeds in cycles of five epochs. In each cycle, the rows of every training file are divided at random into five disjoint parts, and each epoch loads a different part of nearly every file. A row is therefore loaded in a single epoch per cycle, and within that epoch the reweighting can draw it several times. The order of the training examples and the seeds of the dropout masks in each epoch are derived from the seed and the epoch number alone. Every run records, for each epoch, a hash of the identifiers of the jets it read, in order. We use this scheme because in our implementation of Sophon's loading scheme, which reads whole files, the fraction of QCD jets varied between epochs far more than random sampling would.

Network. The network is the ParticleTransformer of weaver-core 0.4.17, the training framework of ParT and Sophon, in the configuration of Sophon's released code: 8 particle-attention blocks, 2 class-attention blocks, embedding dimension 128 and 8 attention heads, followed by a head with a single hidden layer of 512 units. Sophon's paper states 6 particle-attention blocks [44, App. B.1]; we follow its released code. The weights use PyTorch's default initialization, except for the class token, which this implementation draws from a narrow truncated normal distribution. Sophon's training code uses another version of weaver-core, whose default initialization draws the linear layers from a Xavier-uniform distribution with zero biases and rescales the residual projections by depth; we did not reproduce it.

Optimization. Every model is trained for 80 epochs of 10,240,000 jets (8.192×10^8 training examples) with batch size 512 in mixed precision. The optimizer is Ranger: Lookahead [64] with $k = 6$ and $\alpha = 0.5$ over RAdam [47] with $\beta_1 = 0.95$, $\beta_2 = 0.999$, $\epsilon = 10^{-5}$ [44, App. B.1], and no weight decay, as in the ParT training that Sophon follows [54]. The learning rate is 5×10^{-4} for the first 70% of the epochs and then decreases by a constant factor at every epoch from epoch 56, to 5×10^{-6}, 1% of its initial value, in the last epoch.

Checkpoint selection. The validation accuracy is computed on every selected jet of 25 validation files that no run trains on. For the check of Appendix F, the BatchNorm statistics of the selected checkpoint are recomputed on 200,000 training jets.

B Label sets and random partitions

Table 13 lists the levels of the class hierarchy and the groups of the 17-class label set, and Table 14 gives the group of every JetClass-II class at each coarser level, in every random partition and in every random partition with controlled flavor boundaries. Both are generated from the label maps.

Candidate partitions are built as in Section 3.2 from consecutive seeds. Of 6,600 candidates, 27 meet the rule on the realized shares below, and the five partitions are a random sample of these that meets a balance rule fixed before the draw. Each of seven pairs of classes, including those compared in Section 4.2, is merged in two or three of the partitions and kept apart in the others. The realized share of each group lies within 5% of that of its matched group.

In the random partition with a b-quark boundary, the 11 of the 22 four-quark decays $X \to YY \to QQQQ$ that contain a b quark move to another group. To keep the shares exact, the 9 decays of the forms $X \to YY \to QQ\mu\mu$, $X \to YY \to QQ\tau_h\tau_h$ and $X \to \mu\mu$ move the other way. The random partition with a random flavor boundary moves 11 of the same four-quark decays, 6 of them with a b quark, and makes the same compensating move. Its moved decays are drawn under rules fixed before the draw: about as many of them contain a b quark as do not; no pair of them that differ only by exchanging b and c quarks is kept apart; the pairs of the balance rule are merged or kept apart as in the partition with no flavor boundary; and the share of the training examples in each group stays within 1% of that in the partition with a b-quark boundary.

Table 13: The levels of the class hierarchy, finest first: the number of classes, what each level labels, and whether models were pretrained on it. Below, the groups of the 17-class label set, by the group number used in Table 14, with the number of JetClass-II classes each merges. ℓ is an electron or a muon; τ_ℓ a leptonic τ decay. A four-prong group with a neutrino has three visible prongs.

classes	labels	pretrained
188	the JetClass-II classes	yes
162	one QCD class; every resonant class kept	yes
64	each decay topology by its numbers of b and c quarks	no
43	each decay topology by whether it holds a b quark, else a c, else neither	yes
30	each decay topology by whether it holds a b or c quark	no
17	the number of prongs and whether each is a quark or gluon (without distinguishing them), an electron or muon, a τ lepton or a neutrino, without flavor	yes
4	two, three or four visible prongs (a neutrino is not counted), or QCD	no
2	resonance decay or QCD	no
the 17 classes of the 17-class label set		
0	two-prong: two quarks or gluons	10 classes
3	two-prong: $\ell\ell$	2 classes
2	two-prong: τ_h, τ_ℓ	2 classes
1	two-prong: τ_h, τ_h	1 class
9	four-prong: four quarks or gluons	27 classes
11	four-prong: two quarks or gluons, $\ell\ell$	10 classes
12	four-prong: two quarks or gluons, τ_h, τ_ℓ	10 classes
10	four-prong: two quarks or gluons, τ_h, τ_h	5 classes
4	three-prong: three quarks or gluons	29 classes
6	three-prong: two quarks or gluons, ℓ	10 classes
7	three-prong: one quark or gluon, $\ell\ell$	10 classes
8	three-prong: one quark or gluon, τ_h, τ_ℓ	10 classes
5	three-prong: one quark or gluon, τ_h, τ_h	5 classes
13	four-prong: two quarks or gluons, ℓ, ν	12 classes
14	four-prong: two quarks or gluons, τ_ℓ, ν	12 classes
15	four-prong: two quarks or gluons, τ_h, ν	6 classes
16	QCD	27 classes

Table 14: Every JetClass-II class and its group at each coarser level of the class hierarchy (columns headed by the number of classes at that level: 64, 43, 30, 17), in each random partition, numbered as in Table 4, and in each random partition with controlled flavor boundaries: no flavor boundary, a b-quark boundary and a random flavor boundary (Section 3.2). Group numbers are those of the released label maps; classes that share a number in a column share a class in that label set. At the 162-class level every resonant class keeps its own label and the QCD classes share one; at 4 classes the groups are decays with two, three or four visible prongs (a neutrino is not counted) and QCD. Rows are ordered by the 17-class group (Table 13), with a gap between groups. q is a u or d quark; τ_h a hadronic and τ_e, τ_μ leptonic τ decays.

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to bb$	0	0	0	0	0	9	0	9	0	0	0	0
$X \to bc$	1	0	0	0	0	0	0	9	0	0	0	0
$X \to bq$	2	0	0	0	0	0	9	0	0	0	0	0
$X \to cc$	3	1	0	0	0	0	0	0	0	0	0	0
$X \to cs$	4	1	0	0	0	9	9	0	0	0	0	0
$X \to cq$	4	1	0	0	9	0	9	0	0	0	0	0
$X \to ss$	5	2	1	0	9	9	9	9	0	0	0	0
$X \to qq$	5	2	1	0	0	0	9	0	0	0	0	0
$X \to sq$	5	2	1	0	9	0	9	0	0	0	0	0
$X \to gg$	5	2	1	0	3	3	9	13	9	15	15	15
$X \to ee$	8	5	4	3	11	9	13	11	0	0	0	0
$X \to \mu\mu$	8	5	4	3	11	9	9	11	9	11	9	9
$X \to \tau_h\tau_e$	7	4	3	2	15	3	12	12	1	4	4	4
$X \to \tau_h\tau_\mu$	7	4	3	2	4	4	10	10	12	0	0	0
$X \to \tau_h\tau_h$	6	3	2	1	4	10	12	3	4	7	7	7
$X \to YY \to bbbb$	31	21	15	9	0	9	11	0	11	9	11	9
$X \to YY \to bbcc$	32	21	15	9	9	0	11	0	9	9	11	9
$X \to YY \to bbss$	33	21	15	9	9	0	0	0	9	9	11	11
$X \to YY \to bbqq$	33	21	15	9	9	0	11	0	9	9	11	11
$X \to YY \to bbgg$	33	21	15	9	15	15	3	3	13	13	13	13
$X \to YY \to qqqb$	34	21	15	9	0	0	0	0	9	9	11	11
$X \to YY \to qqbs$	34	21	15	9	0	9	11	9	11	9	11	9
$X \to YY \to bbcq$	35	21	15	9	9	0	11	9	11	9	11	11
$X \to YY \to ccbs$	36	21	15	9	0	9	0	0	9	9	11	9
$X \to YY \to ccbq$	36	21	15	9	0	9	11	0	11	9	11	11
$X \to YY \to qqbc$	37	21	15	9	0	9	0	9	9	9	11	11
$X \to YY \to bcsq$	37	21	15	9	0	9	0	0	11	9	11	9
$X \to YY \to cccc$	38	22	15	9	9	9	0	9	9	9	9	9
$X \to YY \to ccss$	39	22	15	9	0	0	0	0	11	9	9	11
$X \to YY \to ccqq$	39	22	15	9	9	9	11	9	11	9	9	11
$X \to YY \to ccgg$	39	22	15	9	10	12	4	13	4	13	13	13
$X \to YY \to ccsq$	39	22	15	9	9	0	0	9	9	9	9	9
$X \to YY \to qqqc$	40	22	15	9	0	9	0	0	11	9	9	11

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to sscq$	40	22	15	9	9	0	0	9	9	9	9	11
$X \to YY \to qqcs$	40	22	15	9	0	9	11	9	11	9	9	9
$X \to YY \to ssss$	41	23	16	9	9	0	11	9	11	9	9	11
$X \to YY \to ssqq$	41	23	16	9	9	0	0	9	9	9	9	9
$X \to YY \to ssgg$	41	23	16	9	3	15	3	7	13	13	13	13
$X \to YY \to qqqq$	41	23	16	9	0	9	11	9	9	9	9	9
$X \to YY \to qqgg$	41	23	16	9	12	10	13	12	12	13	13	13
$X \to YY \to gggg$	41	23	16	9	4	14	12	12	9	4	4	4
$X \to YY \to qqqs$	41	23	16	9	9	0	11	0	11	9	9	9
$X \to YY \to bbee$	45	27	19	11	9	12	0	4	11	3	3	3
$X \to YY \to bb\mu\mu$	45	27	19	11	11	11	0	11	9	11	9	9
$X \to YY \to ccee$	46	28	19	11	11	11	0	11	9	3	3	3
$X \to YY \to cc\mu\mu$	46	28	19	11	11	11	11	4	9	11	9	9
$X \to YY \to ssee$	47	29	20	11	9	13	0	9	9	3	3	3
$X \to YY \to ss\mu\mu$	47	29	20	11	11	13	0	11	13	11	9	9
$X \to YY \to qqee$	47	29	20	11	11	11	11	11	11	3	3	3
$X \to YY \to qq\mu\mu$	47	29	20	11	11	11	0	9	9	11	9	9
$X \to YY \to ggee$	47	29	20	11	9	11	9	9	4	4	4	4
$X \to YY \to gg\mu\mu$	47	29	20	11	10	11	3	11	3	9	9	9
$X \to YY \to bb\tau_h\tau_e$	48	30	21	12	14	13	3	10	12	13	13	13
$X \to YY \to bb\tau_h\tau_\mu$	48	30	21	12	4	9	3	15	7	4	4	4
$X \to YY \to cc\tau_h\tau_e$	49	31	21	12	15	2	10	1	9	13	13	13
$X \to YY \to cc\tau_h\tau_\mu$	49	31	21	12	12	3	4	10	0	4	4	4
$X \to YY \to ss\tau_h\tau_e$	50	32	22	12	15	10	12	6	12	13	13	13
$X \to YY \to ss\tau_h\tau_\mu$	50	32	22	12	12	3	12	6	14	4	4	4
$X \to YY \to qq\tau_h\tau_e$	50	32	22	12	2	4	4	12	1	13	13	13
$X \to YY \to qq\tau_h\tau_\mu$	50	32	22	12	4	0	12	12	4	4	4	4
$X \to YY \to gg\tau_h\tau_e$	50	32	22	12	12	4	10	1	14	1	1	1
$X \to YY \to gg\tau_h\tau_\mu$	50	32	22	12	2	6	2	3	12	14	14	14
$X \to YY \to bb\tau_h\tau_h$	42	24	17	10	4	7	4	14	15	11	9	9
$X \to YY \to cc\tau_h\tau_h$	43	25	17	10	12	13	4	3	15	11	9	9
$X \to YY \to ss\tau_h\tau_h$	44	26	18	10	13	4	15	2	15	11	9	9
$X \to YY \to qq\tau_h\tau_h$	44	26	18	10	13	11	4	9	10	11	9	9
$X \to YY \to gg\tau_h\tau_h$	44	26	18	10	1	11	7	13	10	1	1	1
$X \to YY \to bbb$	9	6	5	4	7	1	3	15	10	10	10	10
$X \to YY \to bbc$	10	6	5	4	4	1	13	10	1	10	10	10
$X \to YY \to bbs$	11	6	5	4	3	4	10	15	6	10	10	10
$X \to YY \to bbq$	11	6	5	4	4	15	10	6	1	10	10	10
$X \to YY \to bbg$	11	6	5	4	6	0	1	14	13	14	14	14
$X \to YY \to ccb$	12	6	5	4	6	12	14	3	2	10	10	10
$X \to YY \to ssb$	13	6	5	4	10	4	13	10	6	10	10	10
$X \to YY \to qqb$	13	6	5	4	9	12	12	10	3	10	10	10
$X \to YY \to ggb$	13	6	5	4	12	7	7	1	13	6	6	6
$X \to YY \to bsq$	13	6	5	4	3	10	12	15	7	10	10	10
$X \to YY \to bcs$	14	6	5	4	1	13	15	3	10	10	10	10

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to bcq$	14	6	5	4	9	12	3	3	10	10	10	10
$X \to YY \to ccc$	15	7	5	4	3	1	10	3	6	10	10	10
$X \to YY \to ccs$	16	7	5	4	7	9	13	12	7	10	10	10
$X \to YY \to ccq$	16	7	5	4	7	4	10	10	7	10	10	10
$X \to YY \to ccg$	16	7	5	4	12	6	14	12	15	14	14	14
$X \to YY \to ssc$	17	7	5	4	10	13	4	12	10	10	10	10
$X \to YY \to qqc$	17	7	5	4	3	1	4	13	15	10	10	10
$X \to YY \to ggc$	17	7	5	4	6	2	1	14	0	6	6	6
$X \to YY \to csq$	17	7	5	4	7	1	15	15	6	10	10	10
$X \to YY \to sss$	18	8	6	4	9	12	9	10	3	10	10	10
$X \to YY \to ssq$	18	8	6	4	3	1	6	10	15	10	10	10
$X \to YY \to ssg$	18	8	6	4	6	10	14	12	14	14	14	14
$X \to YY \to qqs$	18	8	6	4	4	15	10	10	6	10	10	10
$X \to YY \to qqq$	18	8	6	4	4	4	13	3	7	10	10	10
$X \to YY \to qqg$	18	8	6	4	12	7	7	13	0	14	14	14
$X \to YY \to ggs$	18	8	6	4	12	6	14	1	13	6	6	6
$X \to YY \to ggq$	18	8	6	4	1	2	2	14	4	6	6	6
$X \to YY \to ggg$	18	8	6	4	14	14	6	9	2	2	2	2
$X \to YY \to bbe$	22	12	9	6	7	12	10	10	10	14	14	14
$X \to YY \to bb\mu$	22	12	9	6	12	12	12	10	4	0	0	0
$X \to YY \to cce$	23	13	9	6	14	4	1	3	13	14	14	14
$X \to YY \to cc\mu$	23	13	9	6	10	0	14	10	13	0	0	0
$X \to YY \to sse$	24	14	10	6	12	4	14	15	10	14	14	14
$X \to YY \to ss\mu$	24	14	10	6	4	10	6	12	0	0	0	0
$X \to YY \to qqe$	24	14	10	6	7	6	1	15	15	14	14	14
$X \to YY \to qq\mu$	24	14	10	6	6	6	2	10	14	0	0	0
$X \to YY \to gge$	24	14	10	6	3	7	7	12	15	2	2	2
$X \to YY \to gg\mu$	24	14	10	6	6	2	14	1	13	2	2	2
$X \to YY \to bee$	25	15	11	7	14	10	1	10	5	2	2	2
$X \to YY \to b\mu\mu$	25	15	11	7	4	4	6	10	2	4	4	4
$X \to YY \to cee$	26	16	11	7	10	10	6	10	4	2	2	2
$X \to YY \to c\mu\mu$	26	16	11	7	5	10	1	10	15	4	4	4
$X \to YY \to see$	27	17	12	7	14	14	6	9	15	2	2	2
$X \to YY \to qee$	27	17	12	7	10	10	4	10	2	2	2	2
$X \to YY \to gee$	27	17	12	7	10	14	5	9	2	5	5	5
$X \to YY \to s\mu\mu$	27	17	12	7	10	14	5	9	15	4	4	4
$X \to YY \to q\mu\mu$	27	17	12	7	5	10	6	4	15	4	4	4
$X \to YY \to g\mu\mu$	27	17	12	7	14	5	1	9	15	5	5	5
$X \to YY \to b\tau_h\tau_e$	28	18	13	8	8	5	1	5	2	8	8	8
$X \to YY \to b\tau_h\tau_\mu$	28	18	13	8	10	5	8	8	5	8	8	8
$X \to YY \to c\tau_h\tau_e$	29	19	13	8	10	5	1	5	2	8	8	8
$X \to YY \to c\tau_h\tau_\mu$	29	19	13	8	14	14	1	5	2	8	8	8
$X \to YY \to s\tau_h\tau_e$	30	20	14	8	10	14	1	5	5	8	8	8
$X \to YY \to q\tau_h\tau_e$	30	20	14	8	10	8	1	5	5	8	8	8
$X \to YY \to g\tau_h\tau_e$	30	20	14	8	14	8	1	8	8	8	8	8
$X \to YY \to s\tau_h\tau_\mu$	30	20	14	8	14	14	1	5	5	8	8	8

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
$X \to YY \to q\tau_h\tau_\mu$	30	20	14	8	14	5	1	5	8	8	8	8
$X \to YY \to g\tau_h\tau_\mu$	30	20	14	8	8	14	8	5	2	8	8	8
$X \to YY \to b\tau_h\tau_h$	19	9	7	5	8	14	8	8	8	4	4	4
$X \to YY \to c\tau_h\tau_h$	20	10	7	5	14	8	8	5	8	4	4	4
$X \to YY \to s\tau_h\tau_h$	21	11	8	5	8	8	8	8	8	4	4	4
$X \to YY \to q\tau_h\tau_h$	21	11	8	5	10	5	1	8	5	4	4	4
$X \to YY \to g\tau_h\tau_h$	21	11	8	5	8	8	1	5	2	4	4	4
$X \to YY \to bce\nu$	51	33	23	13	9	0	10	11	3	11	11	11
$X \to YY \to bc\mu\nu$	51	33	23	13	9	13	0	4	13	9	9	9
$X \to YY \to bqe\nu$	52	33	23	13	0	4	13	11	12	11	11	11
$X \to YY \to bq\mu\nu$	52	33	23	13	15	13	10	11	3	9	9	9
$X \to YY \to cse\nu$	53	34	23	13	13	15	0	0	3	11	11	11
$X \to YY \to cqe\nu$	53	34	23	13	15	3	15	15	10	11	11	11
$X \to YY \to cs\mu\nu$	53	34	23	13	9	9	15	3	3	9	9	9
$X \to YY \to cq\mu\nu$	53	34	23	13	15	4	10	3	13	9	9	9
$X \to YY \to sqe\nu$	54	35	24	13	13	0	15	15	3	11	11	11
$X \to YY \to qqe\nu$	54	35	24	13	13	4	15	4	3	11	11	11
$X \to YY \to sq\mu\nu$	54	35	24	13	13	13	0	3	3	9	9	9
$X \to YY \to qq\mu\nu$	54	35	24	13	9	11	3	4	13	9	9	9
$X \to YY \to bc\tau_e\nu$	55	36	25	14	7	0	7	14	14	6	6	6
$X \to YY \to bc\tau_\mu\nu$	55	36	25	14	6	12	12	3	10	9	9	9
$X \to YY \to bq\tau_e\nu$	56	36	25	14	14	10	6	1	4	6	6	6
$X \to YY \to bq\tau_\mu\nu$	56	36	25	14	6	7	2	10	4	9	9	9
$X \to YY \to cs\tau_e\nu$	57	37	25	14	4	6	14	12	14	6	6	6
$X \to YY \to cq\tau_e\nu$	57	37	25	14	3	4	14	15	15	6	6	6
$X \to YY \to cs\tau_\mu\nu$	57	37	25	14	7	12	2	13	0	9	9	9
$X \to YY \to cq\tau_\mu\nu$	57	37	25	14	6	6	1	10	15	9	9	9
$X \to YY \to sq\tau_e\nu$	58	38	26	14	1	4	14	12	10	6	6	6
$X \to YY \to qq\tau_e\nu$	58	38	26	14	12	2	14	10	15	6	6	6
$X \to YY \to sq\tau_\mu\nu$	58	38	26	14	10	10	10	10	13	9	9	9
$X \to YY \to qq\tau_\mu\nu$	58	38	26	14	12	0	1	15	13	9	9	9
$X \to YY \to bc\tau_h\nu$	59	39	27	15	0	11	3	9	9	12	12	12
$X \to YY \to bq\tau_h\nu$	60	39	27	15	13	11	15	15	9	12	12	12
$X \to YY \to cs\tau_h\nu$	61	40	27	15	13	11	13	9	3	12	12	12
$X \to YY \to cq\tau_h\nu$	61	40	27	15	13	3	0	0	9	12	12	12
$X \to YY \to sq\tau_h\nu$	62	41	28	15	13	11	9	0	10	12	12	12
$X \to YY \to qq\tau_h\nu$	62	41	28	15	0	15	0	15	12	12	12	12
QCD, *bbccss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbccs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbcs*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbc*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbss*	63	42	29	16	16	16	16	16	16	16	16	16
QCD, *bbs*	63	42	29	16	16	16	16	16	16	16	16	16

	level				random partition					flavor boundary		
JetClass-II class	64	43	30	17	1	2	3	4	5	none	b quark	random
QCD, bb	63	42	29	16	16	16	16	16	16	16	16	16
QCD, $bccss$	63	42	29	16	16	16	16	16	16	16	16	16
QCD, $bccs$	63	42	29	16	16	16	16	16	16	16	16	16
QCD, bcc	63	42	29	16	16	16	16	16	16	16	16	16
QCD, $bcss$	63	42	29	16	16	16	16	16	16	16	16	16
QCD, bcs	63	42	29	16	16	16	16	16	16	16	16	16
QCD, bc	63	42	29	16	16	16	16	16	16	16	16	16
QCD, bss	63	42	29	16	16	16	16	16	16	16	16	16
QCD, bs	63	42	29	16	16	16	16	16	16	16	16	16
QCD, b	63	42	29	16	16	16	16	16	16	16	16	16
QCD, $ccss$	63	42	29	16	16	16	16	16	16	16	16	16
QCD, ccs	63	42	29	16	16	16	16	16	16	16	16	16
QCD, cc	63	42	29	16	16	16	16	16	16	16	16	16
QCD, css	63	42	29	16	16	16	16	16	16	16	16	16
QCD, cs	63	42	29	16	16	16	16	16	16	16	16	16
QCD, c	63	42	29	16	16	16	16	16	16	16	16	16
QCD, ss	63	42	29	16	16	16	16	16	16	16	16	16
QCD, s	63	42	29	16	16	16	16	16	16	16	16	16
QCD, light	63	42	29	16	16	16	16	16	16	16	16	16

C Linear-probe and fine-tuning settings

The inverse regularization strength of the linear probes is chosen from a logarithmic grid between 1×10^{-3} and 100. Each probe is minimized by full-batch L-BFGS in double precision; a fit counts as converged when the largest component of its gradient ends below 1×10^{-5}. Of the probes of pretrained models, 0 did not converge, and of those of the randomly initialized networks 0; 81 of the 780 selected fits chose the largest value of the grid, so for these the best regularization on the validation jets may lie beyond it. The frozen features are stored in half precision. The JetClass-II probes and fine-tuned models are scored on the first 2,000,000 JetClass-II test jets, and their macro AUCs on every fourth of these, 500,000 jets, to bound the cost of 188 ROC curves per model; the binary probes of Table 4 use every jet of both sides among the 2,000,000. These jets come from the first files of the test split, which alternate between the decay families file by file, so they hold 20.5% QCD and 38.7% two-prong jets, against 13.5% and 15.4% in the largest training subset. A one-versus-rest AUC compares the jets of a class with all other jets and so depends on this mix only through the composition of the other jets. The rarest class has 11 jets in the largest training subset and 10 among the jets of the macro AUC, so the AUCs of the rarest classes are

noisy; the QCD pairs of Table 4, with many more jets, measure the QCD subdivision more precisely. JetClass models are scored on 2,000,000 test jets, the first test files of each class with equal numbers of jets per class, and their macro AUC on 500,000 of these; ParT used all of the JetClass test set [54]. The probes choose their regularization on 200,000 JetClass validation jets. The binary probes of Table 4 choose theirs on the validation jets of both sides of the pair, between 302 and 2,108 jets, so the selected value is noisy and its effect is part of the spread between pretraining runs. The smallest JetClass-II training subset contains jets of 160 of the 188 classes, with a median of 4 jets per class, and the next contains 183. In a linear probe, a class absent from training receives probability zero and the AUC of a random guess, which raises $1 - \text{AUC}$ at these sizes for every model alike; Table 16 also gives the average over the classes present in training.

Every fine-tuning setting below, except the schedule of the head's learning rate, is read from the record that each fine-tuning run writes when it starts. The pretrained weights are trained at a constant learning rate of 10^{-4} and the new head at 50 times that, the rates ParT uses for fine-tuning [54, Sec. 5.2], with weight decay 0.01 and batches of 512 jets, with the Ranger optimizer of pretraining in mixed precision. The new head is the whole classification head, its hidden layer included, with dropout 0.1 after the hidden layer. The hidden layer is initialized afresh as well, so that fine-tuning starts from the representation the probes evaluate, the class token, with the same head for every label set; a pretrained hidden layer was trained to serve its own label set's outputs. On top and quark/gluon tagging both rates stay constant; on JetClass-II and JetClass the learning rate of the head falls to 1% of its value over the last 30% of the epochs, the default schedule of the training framework. On JetClass-II and JetClass, a model is fine-tuned for 50, 50, 30 and 10 epochs at 10^3, 10^4, 10^5 and 10^6 training jets, and on top and quark/gluon tagging for 20 epochs at every size. An epoch draws at least 10,000 training jets, so at 10^3 jets it passes over the training set several times; on JetClass-II and JetClass an epoch is 19, 19, 195 and 1,953 steps at these sizes. The pretrained models and the network trained from scratch get the same number of steps, so they are compared at equal training compute. This budget may be short: ParT, trained from scratch on 2% of JetClass, was trained for 100 k steps and was converged by then [54, Sec. 5.1], more steps than our largest JetClass subset receives. Section 4.5 therefore repeats the largest JetClass size with longer training. The epoch with the highest validation accuracy is selected on the same 20,480 jets in every epoch on JetClass-II, and on jets drawn anew in each epoch on the other tasks: 20,000 on JetClass, and on top and quark/gluon tagging 20,000

at 10^3 and 10^4 training jets and 200,000 at the larger sizes. The new head is initialized afresh for every fine-tuning run, and the pretrained weights are loaded from the selected checkpoint, whose hash every run records. Each fine-tuning run is evaluated at its epoch of highest validation accuracy, the earliest in case of a tie; a run that resumed after an interruption takes that epoch over all of its epochs.

The reference trained from scratch follows ParT's recipe for training from scratch [54, Secs. 5.1 and 5.2]: a learning rate of 10^{-3} for every weight, held constant and then lowered to 1% of its value over the last 30% of the epochs, with weight decay 0.01 on top and quark/gluon tagging and none on JetClass-II and JetClass, and the same epochs and batches as fine-tuning. All fine-tuning, including the reference trained from scratch, ran on V100 GPUs.

For $X \to b\bar{b}$ and $X \to bs$, Sophon's transfer classifier is trained on 2.5 million jets with batch size 1024 and a constant learning rate of $5 \times 10{-}4$ [44, App. B.5]. Sophon does not state its optimizer or regularization; we use the optimizer of pretraining, dropout in the hidden layer, no weight decay and several seeds. When our training sample is smaller than these 2.5 million jets, it is repeated: **[samples not in]** times for $X \to b\bar{b}$ and **[samples not in]** times for $X \to bs$. The network trained from scratch on the same jets follows Sophon's reference [44, App. B.3] in its number of epochs, 50, and in the number of jets per epoch, but it is the Sophon network on our inputs, rather than ParT, and selects its epoch on our validation jets.

D Compute

The pretraining runs started on a single GPU type. When that type became unavailable partway through the study, the runs still queued or interrupted were allowed to use any GPU type with enough memory, and a resumed run could continue on another type. Runs resumed on another type read the same training examples in the same order as the other runs of their seed, which each epoch's record of the examples it read confirms; the floating-point rounding and the dropout masks do change with the type, and the effect of the type on a trained model was not measured. Table 15 gives the GPU types and hours of every pretraining run, read from the record that each epoch keeps of its device. The hours count the training and validation passes of the epochs of the final runs and leave out restarted epochs and the computation of the BatchNorm statistics. The runs used L4, RTX 3090 and V100 GPUs, and each level of the hierarchy spent between 56% and 71% of

Table 15: GPU types and GPU-hours of every pretraining run, summed over the training and validation passes of its completed epochs as each epoch recorded them.

run	GPU types (hours)	GPU-hours
188 classes, seed 1	V100 (135)	135
188 classes, seed 2	V100 (145)	145
188 classes, seed 3	RTX 3090 (116)	116
162 classes, seed 1	V100 (75), RTX 3090 (54)	129
162 classes, seed 2	V100 (86), RTX 3090 (40)	126
162 classes, seed 3	V100 (123), L4 (39)	162
43 classes, seed 1	RTX 3090 (113)	113
43 classes, seed 2	V100 (101), RTX 3090 (24)	125
43 classes, seed 3	V100 (107), RTX 3090 (20)	127
17 classes, seed 1	RTX 3090 (108), L4 (3)	111
17 classes, seed 2	V100 (133)	133
17 classes, seed 3	V100 (101), L4 (72)	174
162 classes, mass regression, seed 1	V100 (129), RTX 3090 (26)	155
162 classes, mass regression, seed 2	V100 (130), RTX 3090 (15)	145
17 classes, mass regression, seed 1	V100 (126), RTX 3090 (16)	142
17 classes, mass regression, seed 2	RTX 3090 (101)	101
random partition 1, seed 1	RTX 3090 (78), V100 (38)	116
random partition 2, seed 2	RTX 3090 (110)	110
random partition 3, seed 1	RTX 3090 (109)	109
random partition 4, seed 2	RTX 3090 (107)	107
random partition 5, seed 2	RTX 3090 (98)	98
random partition, no flavor boundary, seed 1	RTX 3090 (99)	99
random partition, no flavor boundary, seed 2	RTX 3090 (130)	130
random partition, b-quark boundary, seed 1	V100 (98), RTX 3090 (24)	122
random partition, b-quark boundary, seed 2	RTX 3090 (137)	137
random partition, random flavor boundary, seed 1	RTX 3090 (125)	125
random partition, random flavor boundary, seed 2	V100 (135), RTX 3090 (7)	142

its GPU-hours on the V100, the type these levels used most.

E Linear probes against the training-set size

Table 16 gives the values shown in Fig. 3.

F Linear probes with BatchNorm statistics recomputed

Every linear probe of Table 2 was repeated on the selected checkpoint with its BatchNorm statistics recomputed on 200,000 training jets. For every pretrained run and every task of that table, $1 - \mathrm{AUC}$ changes by at most

0.5%, far less than the differences between label sets, so no result depends on the BatchNorm statistics accumulated in training.

G Test-sample uncertainty

Every model is scored on the same test jets, so the finite test sample moves all of them together, and much of its effect cancels in a ratio between models. Its size is estimated with a bootstrap: the test jets are resampled with replacement within each class [**bootstrap**] times, every model is scored on each resample, and the standard deviation of the logarithm of a ratio over the resamples is its relative uncertainty from the test sample. Table 17 compares it with the spread over pretraining runs for the ratios that Section 4 quotes. For a label set against the 188 classes, the test sample contributes at most [**bootstrap**]% for the linear probes and [**bootstrap**]% after fine-tuning.

Table 16: Linear probes against the training-set size: $(1-\mathrm{AUC})\times 10^3$, lower is better; the JetClass-II macro AUC is also given over the classes present in the training subset and over its QCD classes; column headers are numbers of training jets. Mean ± standard deviation over three pretraining runs. *: a probe that did not converge.

JetClass-II	1,000	10,000	100,000	1,000,000
188 classes	121.9 ± 0.7	49.2 ± 0.5	28.60 ± 0.14	24.01 ± 0.04
162 classes	123.9 ± 0.8	50.57 ± 0.32	30.7 ± 0.4	25.19 ± 0.31
43 classes	130.5 ± 2.1	55.3 ± 1.1	33.4 ± 0.8	28.1 ± 0.6
17 classes	158.3 ± 2.2	69.5 ± 1.4	44.3 ± 0.4	36.9 ± 0.6
randomly initialized network	410 ± 6	306 ± 4	221.5 ± 2.1	194.2 ± 1.8
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

JetClass-II, classes present in training	1,000	10,000	100,000	1,000,000
188 classes	55.7 ± 0.9	36.9 ± 0.5	28.60 ± 0.14	24.01 ± 0.04
162 classes	58.0 ± 0.9	38.29 ± 0.33	30.7 ± 0.4	25.19 ± 0.31
43 classes	65.9 ± 2.5	43.1 ± 1.1	33.4 ± 0.8	28.1 ± 0.6
17 classes	98.5 ± 2.5	57.8 ± 1.4	44.3 ± 0.4	36.9 ± 0.6
randomly initialized network	395 ± 7	301 ± 5	221.5 ± 2.1	194.2 ± 1.8
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

QCD	1,000	10,000	100,000	1,000,000
188 classes	351.8 ± 1.0	172.0 ± 2.3	75.3 ± 2.5	53.5 ± 0.4
162 classes	355.3 ± 1.6	181.9 ± 1.5	88.4 ± 3.4	61.7 ± 2.0
43 classes	357.5 ± 3.5	189.5 ± 3.3	91 ± 5	66 ± 4
17 classes	376 ± 6	222 ± 6	125.5 ± 1.0	93 ± 4
randomly initialized network	473.4 ± 3.0	408.2 ± 2.2	331.1 ± 3.2	273 ± 4
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

JetClass	1,000	10,000	100,000	1,000,000
188 classes	49.9 ± 2.1	36.6 ± 0.4	32.4 ± 0.6	31.9 ± 0.6
162 classes	49.6 ± 2.5	36.3 ± 0.4	32.27 ± 0.20	31.76 ± 0.21
43 classes	49.7 ± 0.4	36.64 ± 0.23	32.43 ± 0.34	32.0 ± 0.4
17 classes	55.1 ± 1.3	39.8 ± 1.1	35.5 ± 0.8	35.0 ± 0.8
randomly initialized network	244.7 ± 3.3	185.1 ± 1.1	173.6 ± 1.1	172.2 ± 1.0
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

top	1,000	10,000	100,000	1,200,000
188 classes	77 ± 9	61 ± 8	59 ± 8	59 ± 8
162 classes	69 ± 6	54 ± 6	52 ± 6	52 ± 6
43 classes	78 ± 10	65 ± 9	62 ± 9	62 ± 9
17 classes	72.9 ± 3.0	60 ± 5	58 ± 4	58 ± 4
randomly initialized network	73.3 ± 1.1	62.7 ± 0.7	61.2 ± 0.5	61.1 ± 0.5
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

q/g	1,000	10,000	100,000	1,600,000
188 classes	137 ± 6	125 ± 6	122 ± 6	122 ± 6
162 classes	149 ± 12	134 ± 10	131 ± 9	131 ± 9
43 classes	145 ± 7	133 ± 6	130 ± 6	129 ± 6
17 classes	151 ± 12	138 ± 11	134 ± 11	134 ± 11
randomly initialized network	165 ± 5	143.6 ± 1.8	139.4 ± 1.6	138.8 ± 1.6
randomly initialized network, BatchNorm recomputed	[no run yet]	[no run yet]	[no run yet]	[no run yet]

Table 17: Relative uncertainty of the ratios of $(1 - \mathrm{AUC})$ at the largest training set of each task, in per cent: the standard deviation of the logarithm of the ratio over the pretraining runs, then that from the finite test sample (a bootstrap of the test jets with **[bootstrap]** resamples, averaged over the runs). The baseline is the randomly initialized network of the same seed for the linear probes and the network trained from scratch on the same subset for fine-tuning. The macro AUC is used for JetClass-II and JetClass.

[no result yet]

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